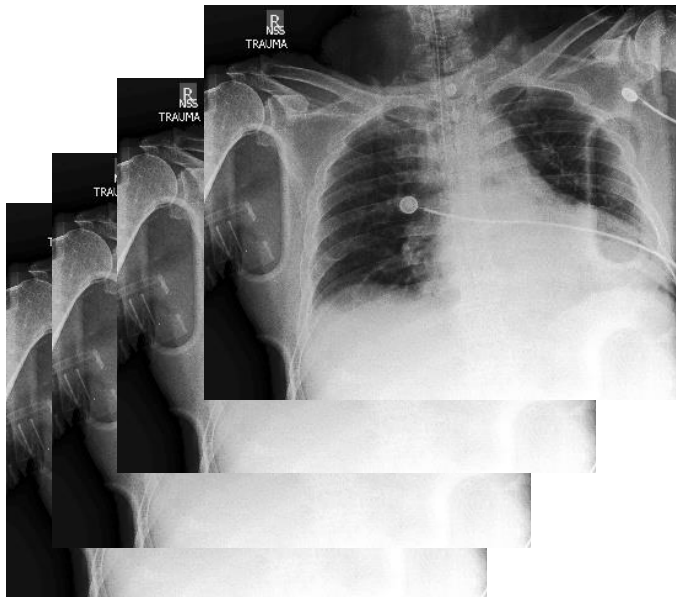
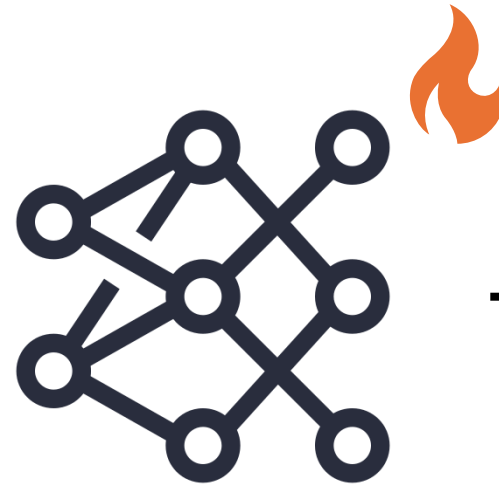
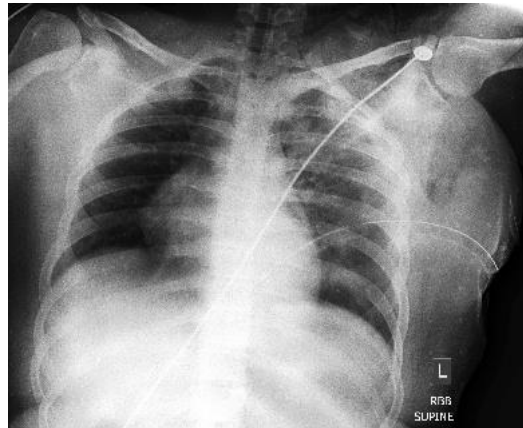




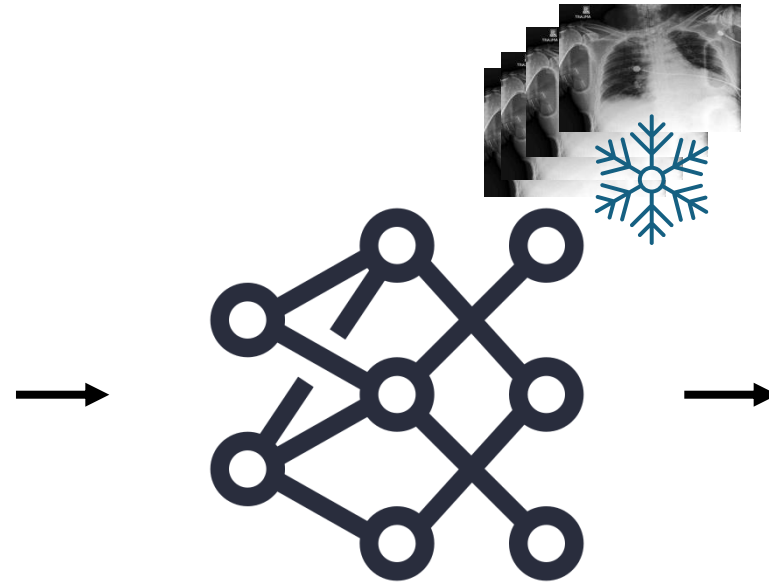
CheXpert



Cardiomegaly
(enlargement of the heart)



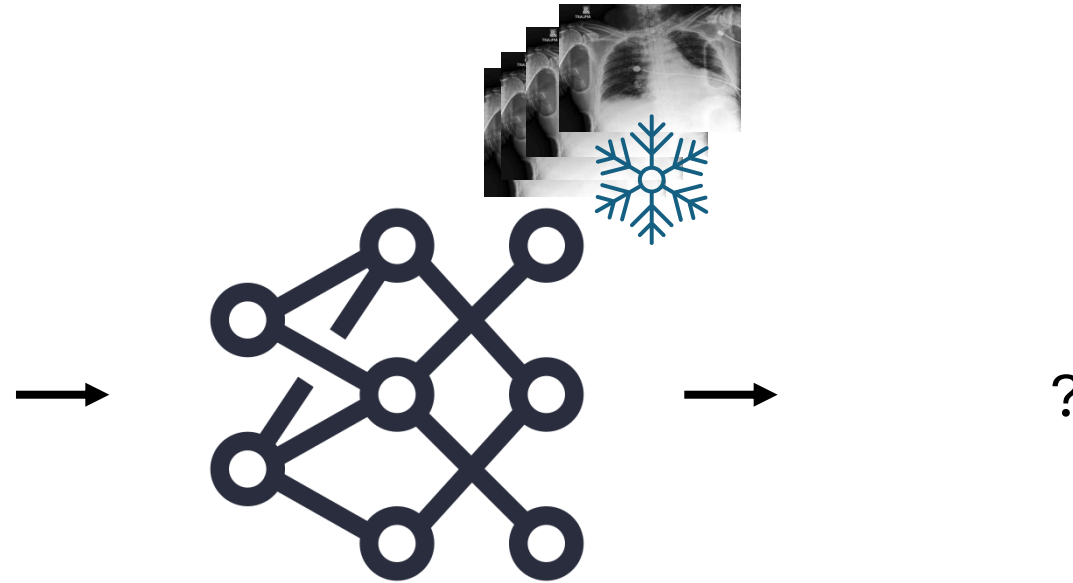
CheXpert



Cardiomegaly
(enlargement of the heart)



Multi-Class Knee Osteoporosis X-Ray Dataset



Out-of-distribution:

A Meta View

Anna M. Wundram

Scope of this lecture

Chapter 1	Data Shifts in Theory
Chapter 2	Data Shifts in Reality
Chapter 3	Fields Addressing Data Shifts
Chapter 4	OOD Detection • Methods • Limitations
Chapter 5	Outlook
Chapter 6	Hands-on

In distribution (ID)

Let x be the inputs to the model (the covariates) and y be the outputs to be predicted (the labels).

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$$p(x, y) = p(x)p(y|x)$$

X causes Y

drug dosage $\rightarrow$ blood pressure

In distribution (ID)

Let x be the inputs to the model (the covariates) and y be the outputs to be predicted (the labels).

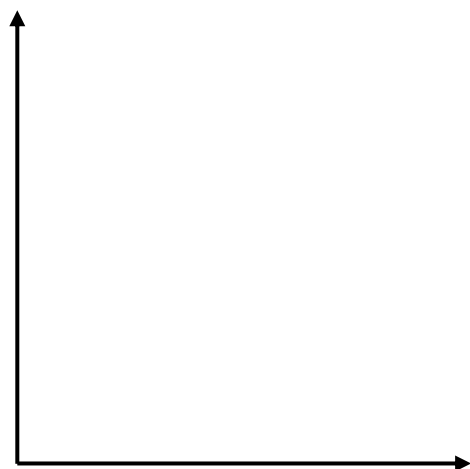
$$p(x, y) = p(x)p(y|x) = p(y)p(x|y)$$

X causes Y

Y causes X

disease → symptoms

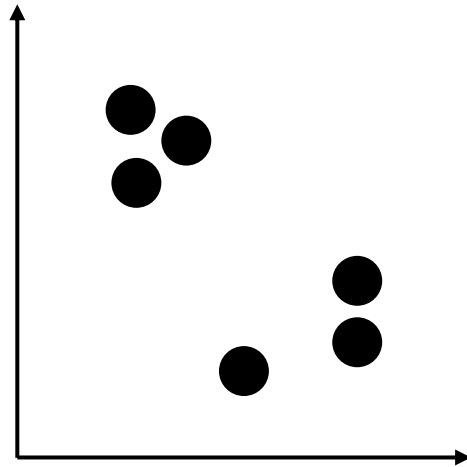
In distribution (ID)



$$p(x, y) = p(x)p(y|x) = p(y)p(x|y)$$

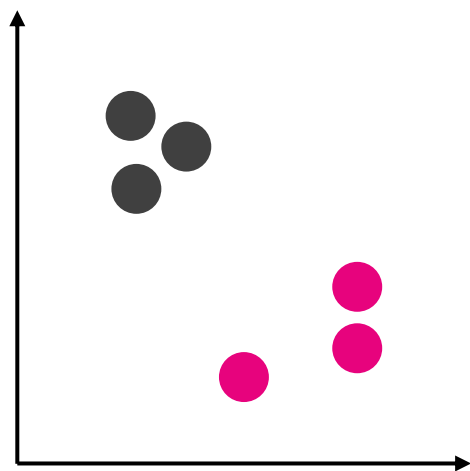
X causes Y Y causes X

In distribution (ID)



$$p(x, y) = \underset{\text{X causes Y}}{\boldsymbol{p}(\boldsymbol{x})} p(y|x) = p(y) \underset{\text{Y causes X}}{p(x|y)}$$

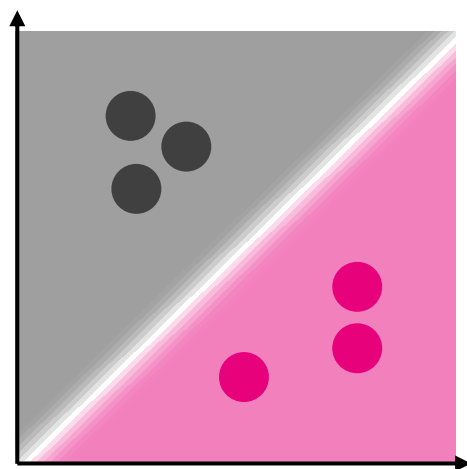
In distribution (ID)



$$p(x, y) = p(x)p(y|x) = \mathbf{p(y)}p(x|y)$$

X causes Y Y causes X

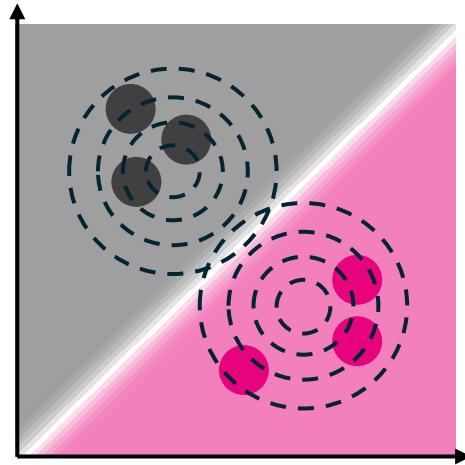
In distribution (ID)



$$p(x, y) = p(x) \mathbf{p(y|x)} = p(y) p(x|y)$$

X causes Y Y causes X

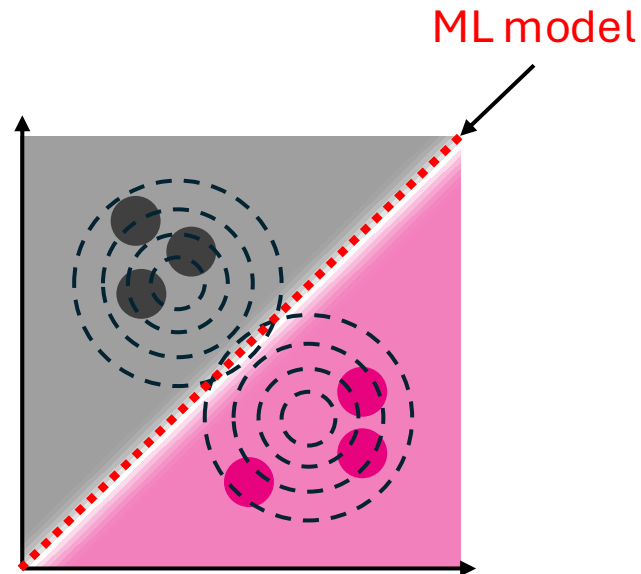
In distribution (ID)



$$p(x, y) = p(x)p(y|x) = p(y)\mathbf{p}(x|y)$$

X causes Y Y causes X

In distribution (ID)



$$p(x, y) = p(x) \textcolor{red}{p(y|x)} = p(y) \textcolor{green}{p(x|y)}$$

X causes Y Y causes X

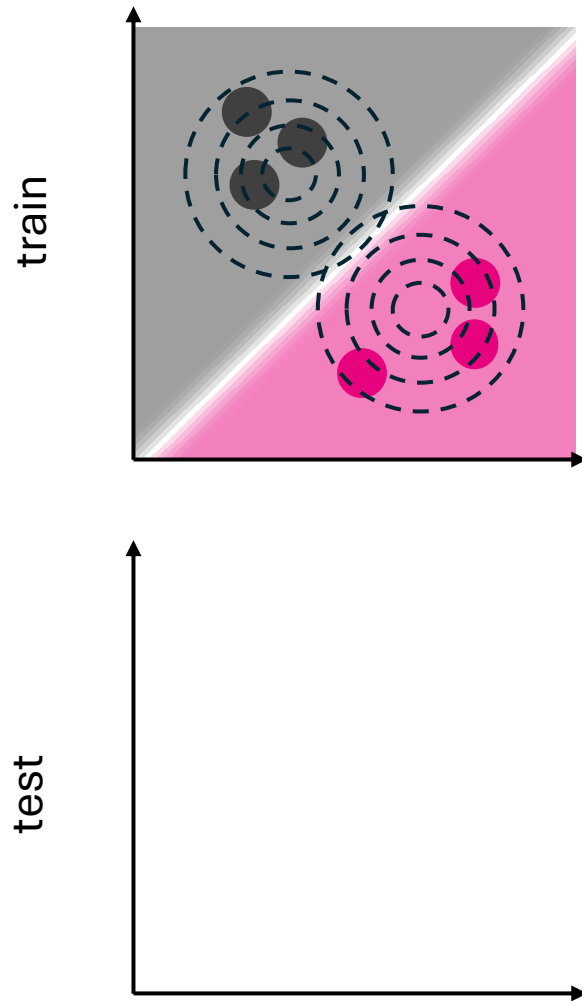
What is out of distribution (OOD)?

$$p_{train}(x, y) \neq p_{test}(x, y)$$

Why?

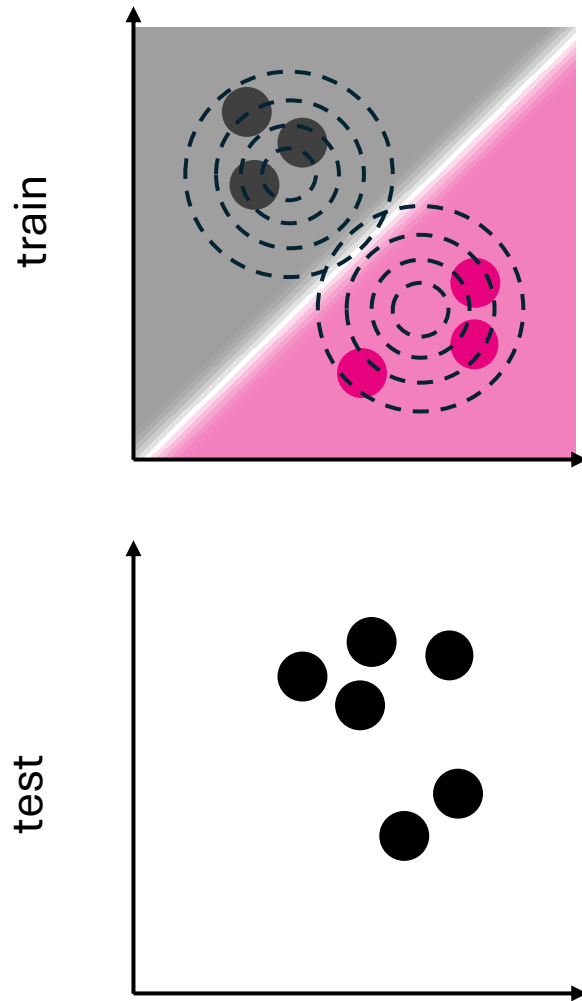
The joint distribution has shifted

Covariate Shift



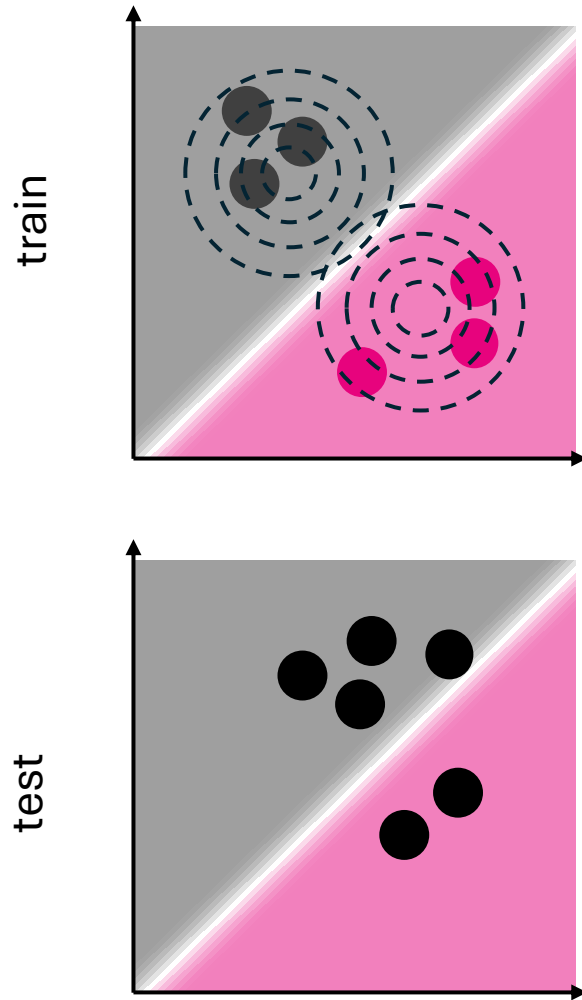
$$p(\mathbf{x})p(y|\mathbf{x}) \rightarrow q(\mathbf{x})p(y|\mathbf{x})$$

Covariate Shift



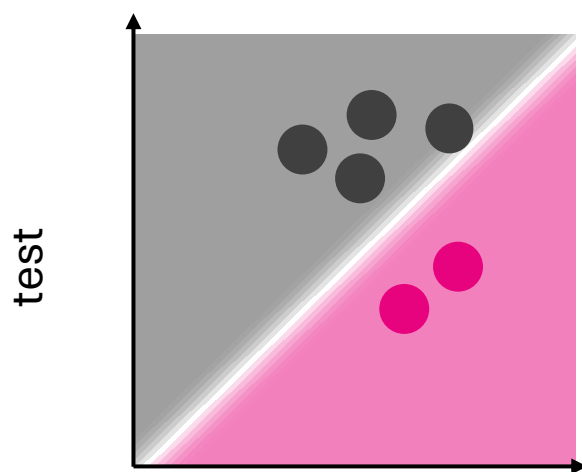
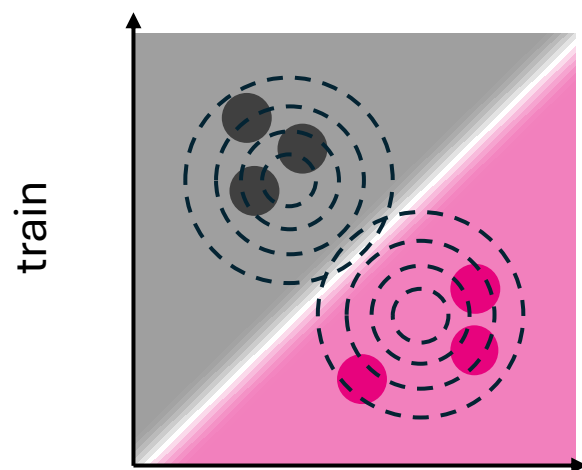
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Covariate Shift



$$p(\mathbf{x})p(y|\mathbf{x}) \rightarrow q(\mathbf{x})p(y|\mathbf{x})$$

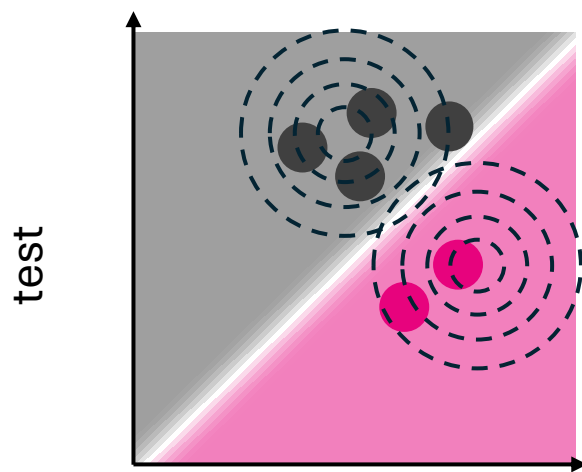
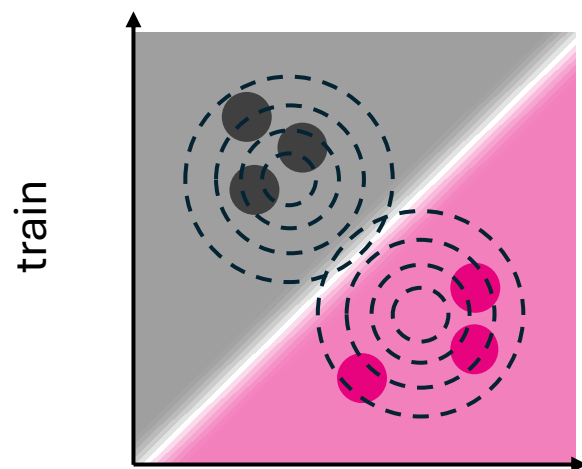
Covariate Shift



$$p(\mathbf{x})p(y|\mathbf{x}) \rightarrow q(\mathbf{x})p(y|\mathbf{x})$$

$$q(y) = \int p(y|\mathbf{x}) \cdot q(\mathbf{x}) d\mathbf{x}$$

Covariate Shift

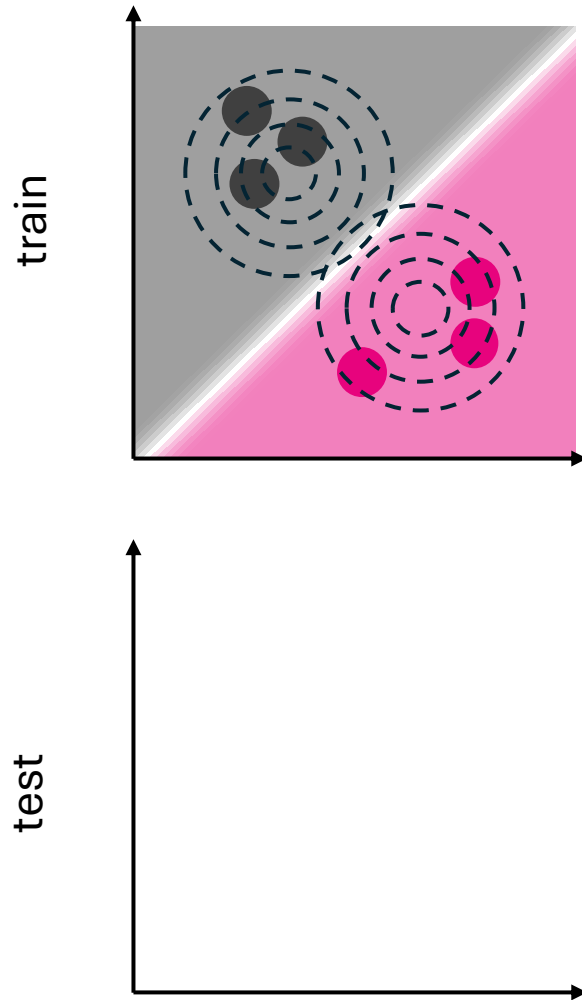


$$p(\mathbf{x})p(y|\mathbf{x}) \rightarrow \mathbf{q}(\mathbf{x})p(y|\mathbf{x})$$

$$q(y) = \int p(y|\mathbf{x}) \cdot q(\mathbf{x}) d\mathbf{x}$$

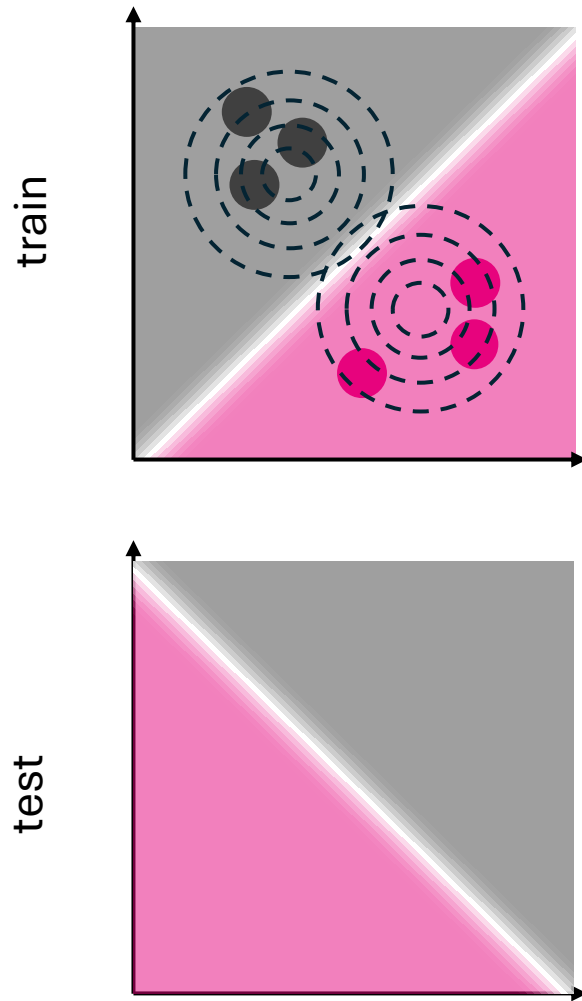
$$q(\mathbf{x}|\mathbf{y}) = \frac{p(y|\mathbf{x})q(\mathbf{x})}{q(y)}$$

Concept Shift



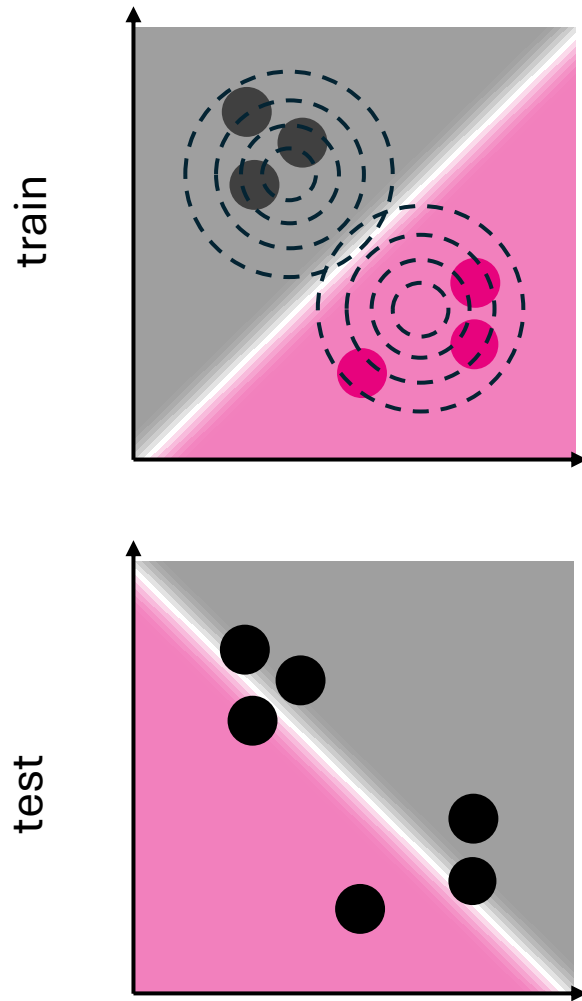
$$p(x)p(y|x) \rightarrow p(x)q(y|x)$$

Concept Shift



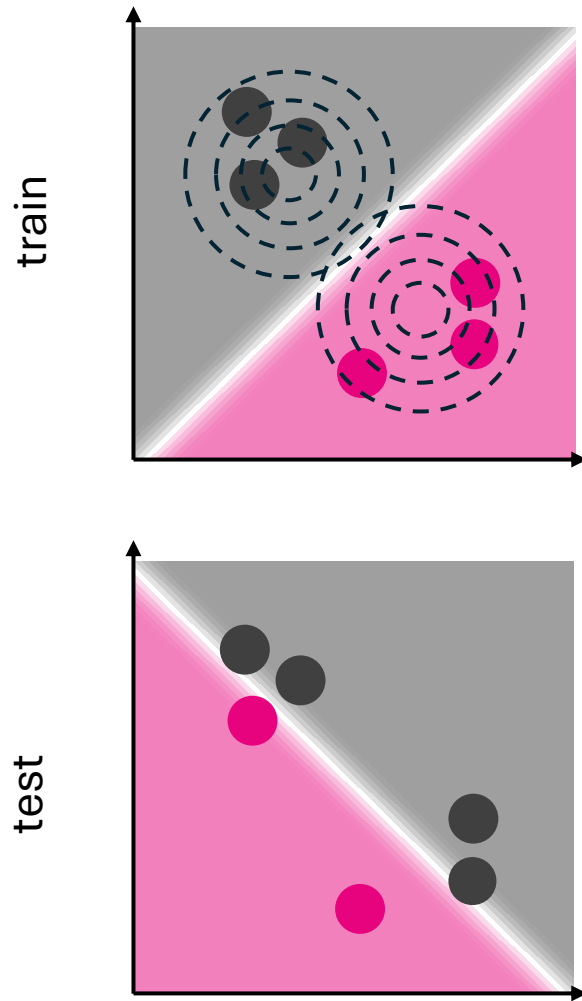
$$p(x)p(y|x) \rightarrow p(x)q(y|x)$$

Concept Shift



$$p(x)p(y|x) \rightarrow p(x)q(y|x)$$

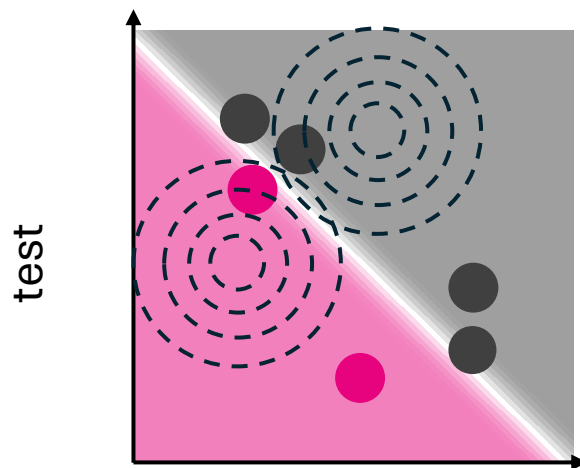
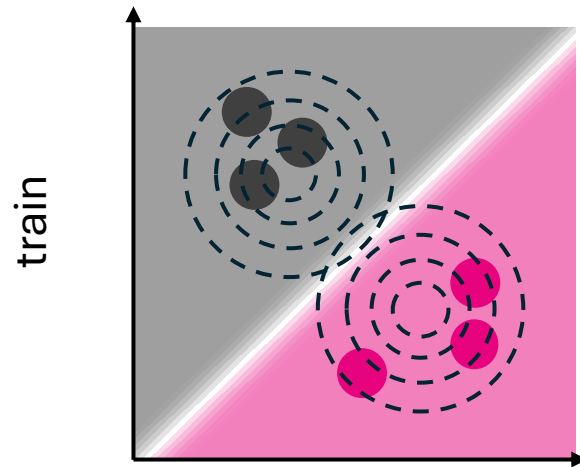
Concept Shift



$$p(x)\mathbf{p}(\mathbf{y}|\mathbf{x}) \rightarrow p(x)\mathbf{q}(\mathbf{y}|\mathbf{x})$$

$$q(y) = \int q(y|x) \cdot p(x) dx$$

Concept Shift

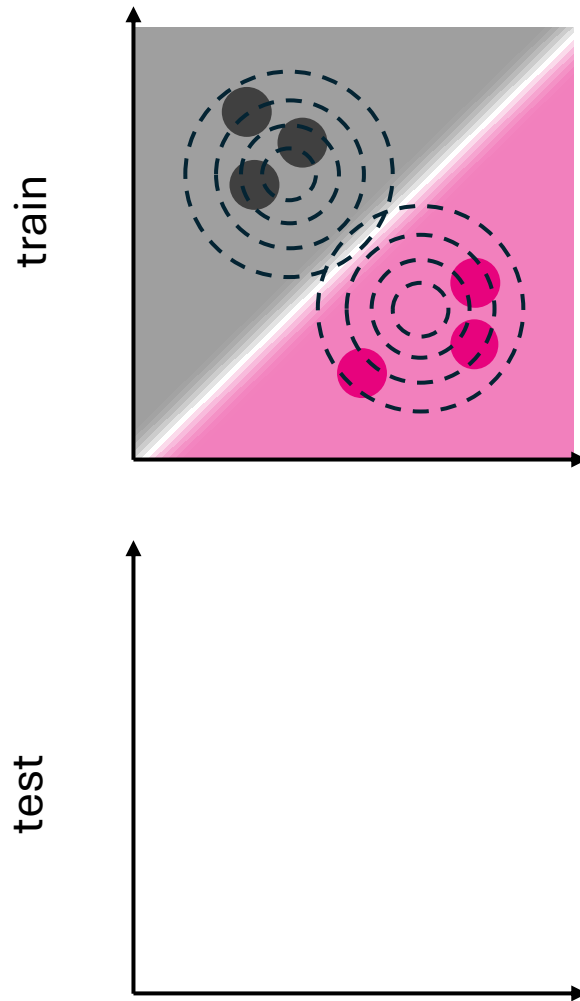


$$p(x)\mathbf{p}(\mathbf{y}|\mathbf{x}) \rightarrow p(x)\mathbf{q}(\mathbf{y}|\mathbf{x})$$

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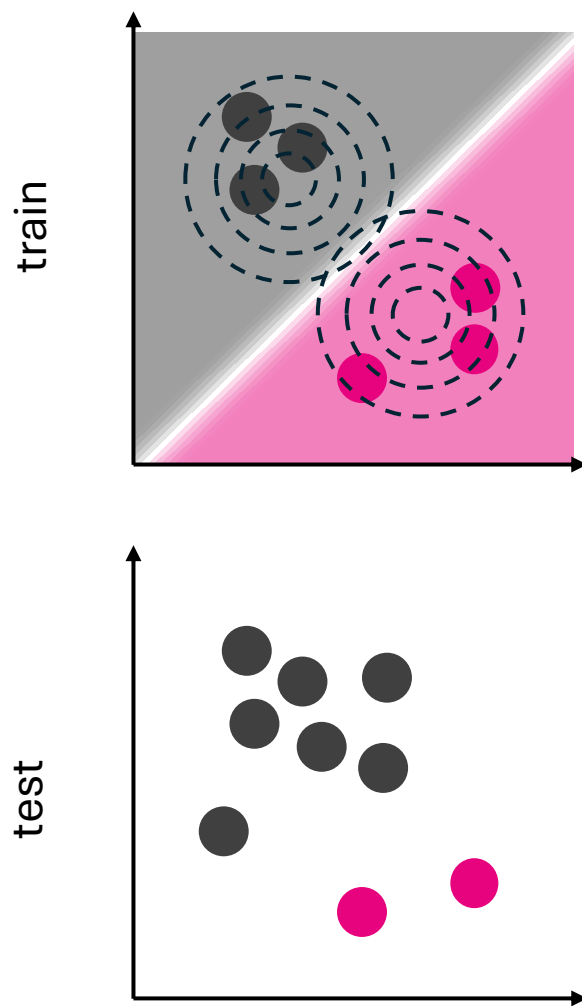
$$q(x|y) = \frac{q(y|x)p(x)}{q(y)}$$

Label shift



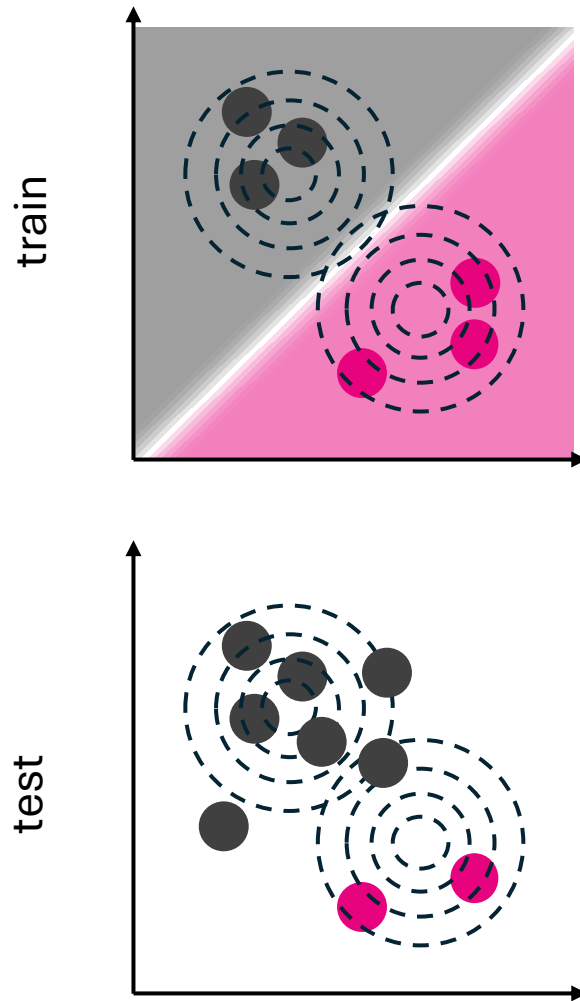
$$p(\mathbf{y})p(x|\mathbf{y}) \rightarrow \mathbf{q}(\mathbf{y})p(x|\mathbf{y})$$

Label shift



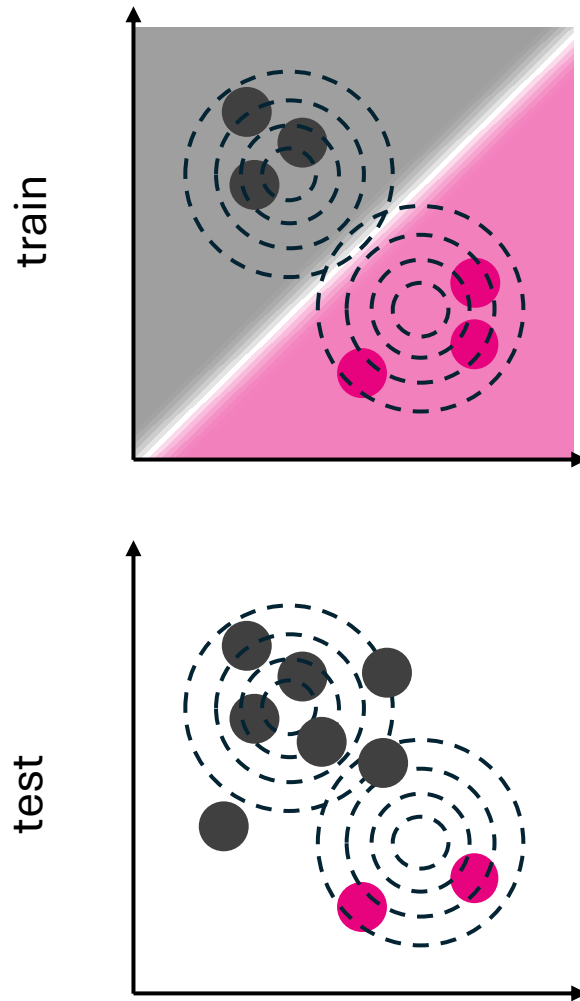
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Label shift



$$p(\mathbf{y})p(x|\mathbf{y}) \rightarrow \mathbf{q}(\mathbf{y})p(x|\mathbf{y})$$

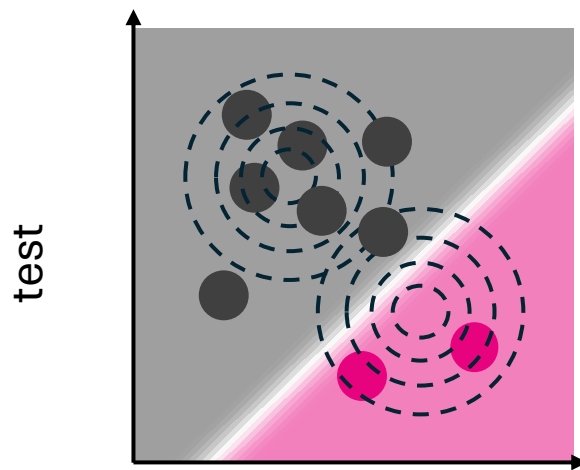
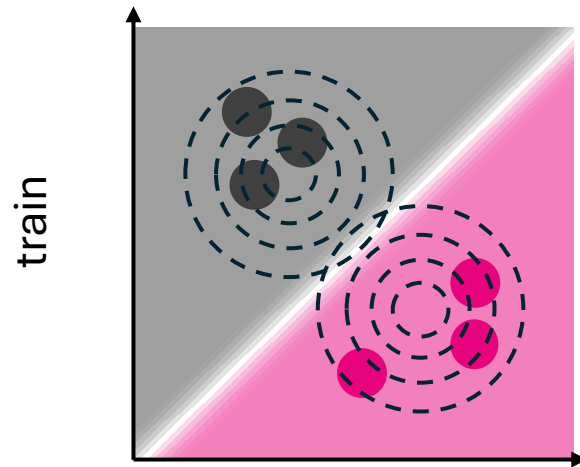
Label shift



$$p(\mathbf{y})p(\mathbf{x}|\mathbf{y}) \rightarrow \mathbf{q}(\mathbf{y})p(\mathbf{x}|\mathbf{y})$$

$$q(\mathbf{x}) = \int p(\mathbf{x}|\mathbf{y}) \cdot q(\mathbf{y}) d\mathbf{y}$$

Label shift

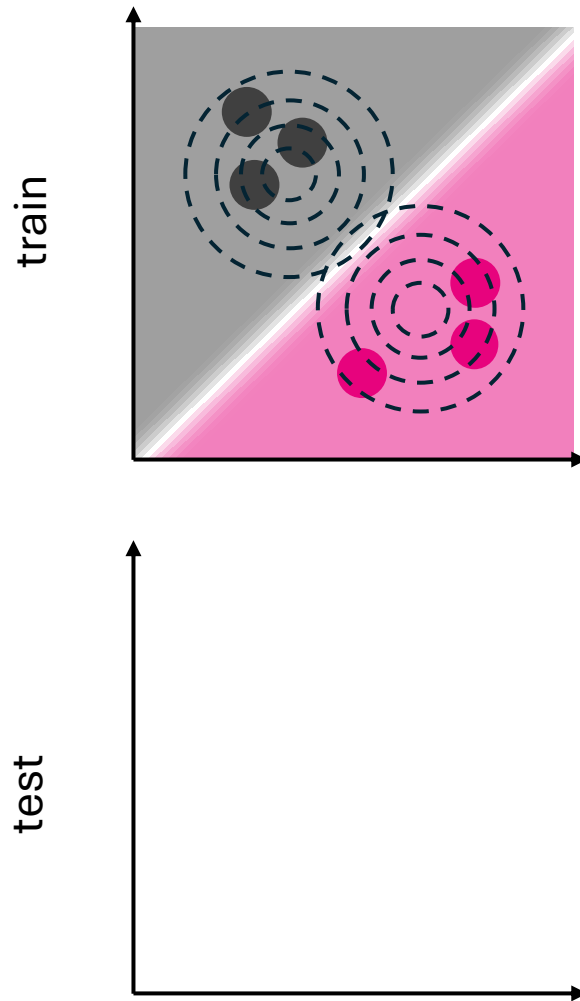


$$p(\mathbf{y})p(\mathbf{x}|\mathbf{y}) \rightarrow \mathbf{q}(\mathbf{y})p(\mathbf{x}|\mathbf{y})$$

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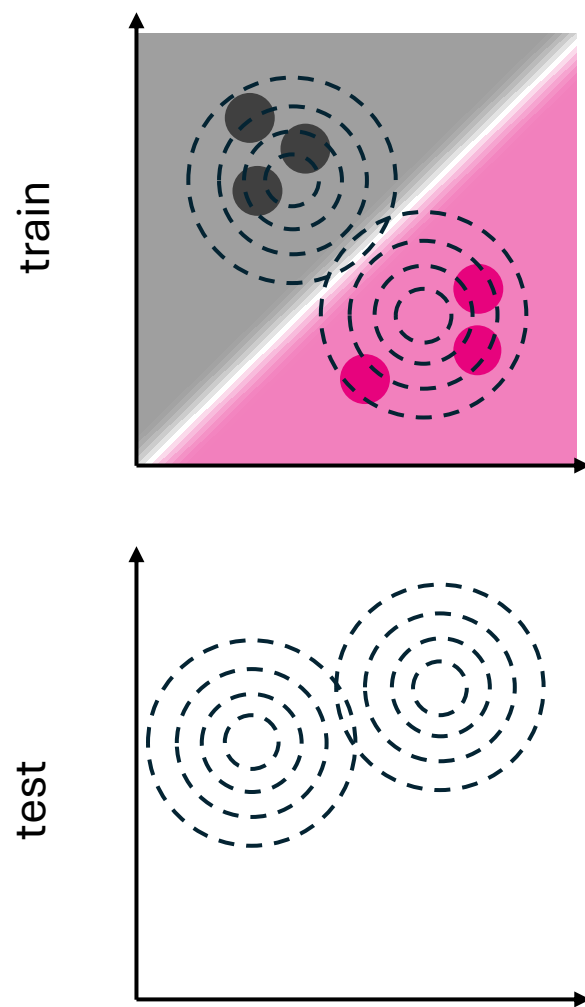
$$q(\mathbf{y}|\mathbf{x}) = \frac{p(\mathbf{x}|\mathbf{y})q(\mathbf{y})}{q(\mathbf{x})}$$

Manifestation shift



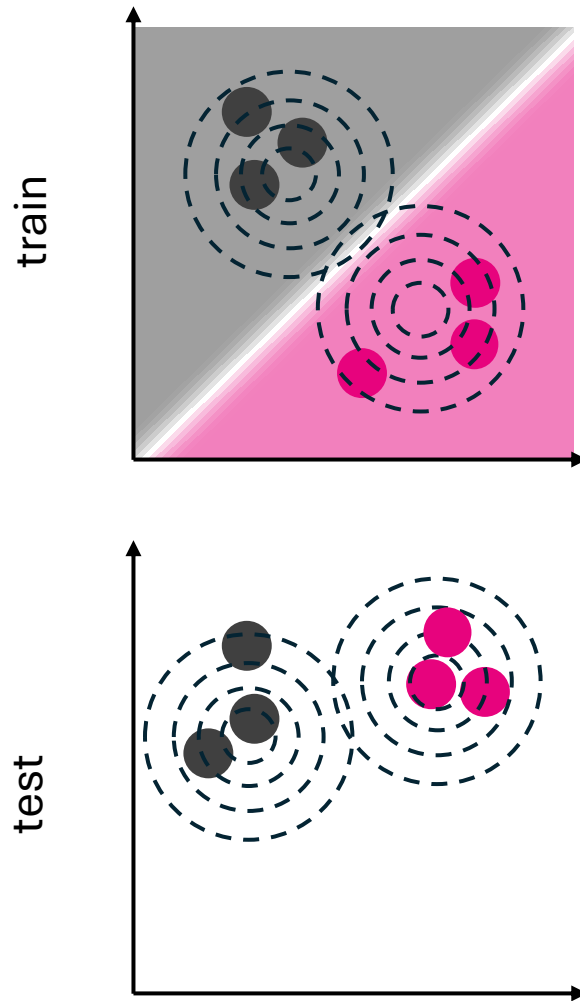
$$p(y)\mathbf{p}(\mathbf{x}|\mathbf{y}) \rightarrow p(y)\mathbf{q}(\mathbf{x}|\mathbf{y})$$

Manifestation shift



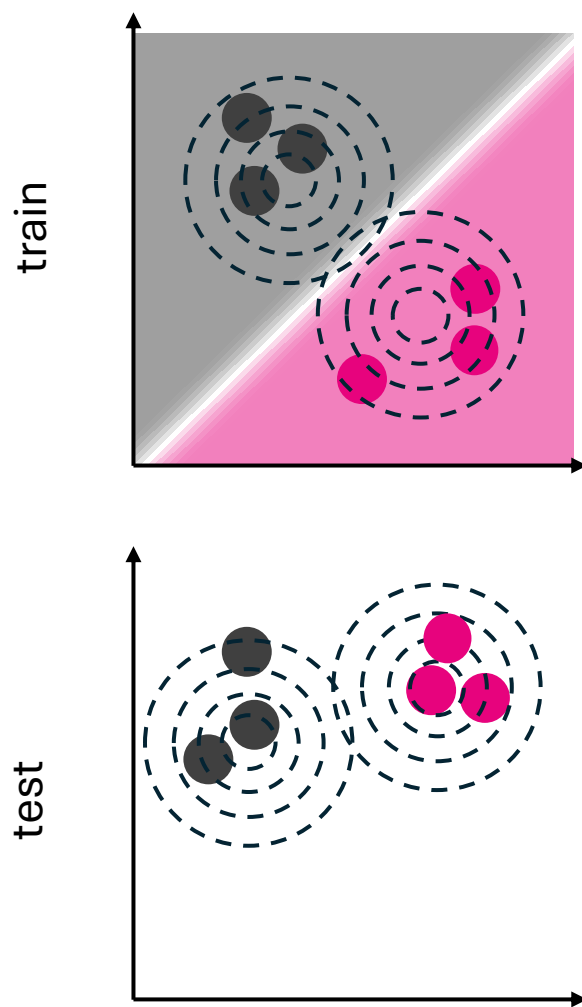
$$p(y)\mathbf{p}(x|y) \rightarrow p(y)\mathbf{q}(x|y)$$

Manifestation shift



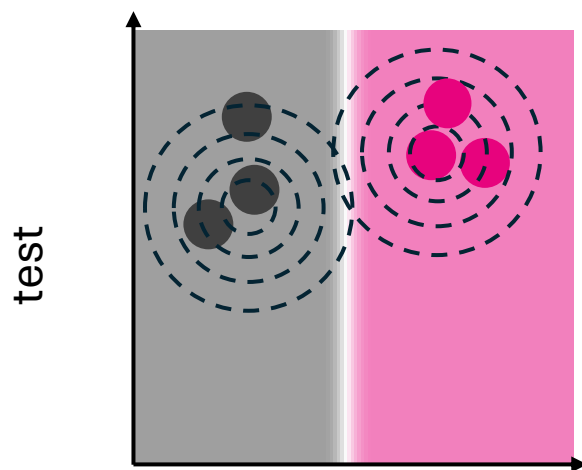
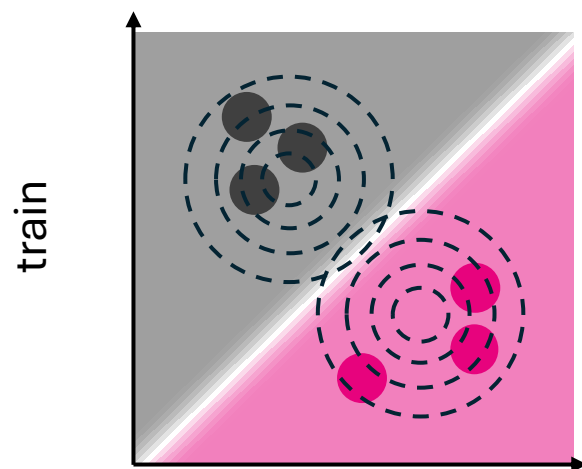
$$p(y)\mathbf{p}(x|y) \rightarrow p(y)\mathbf{q}(x|y)$$

Manifestation shift



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Manifestation shift

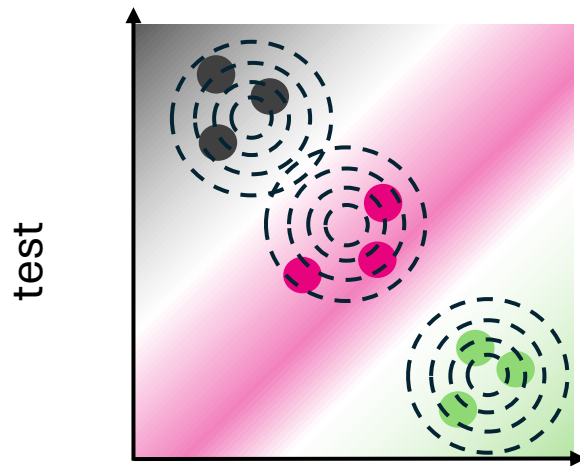
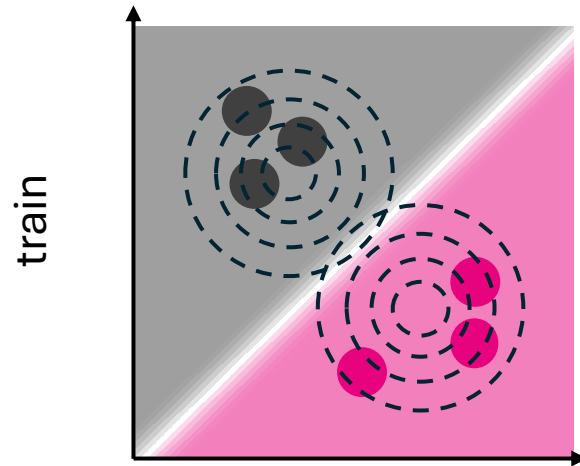


$$p(y)\mathbf{p}(\mathbf{x}|\mathbf{y}) \rightarrow p(y)\mathbf{q}(\mathbf{x}|\mathbf{y})$$

$$q(x) = \int q(x|y) \cdot p(y) dy$$

$$q(y|x) = \frac{q(x|y)p(y)}{q(x)}$$

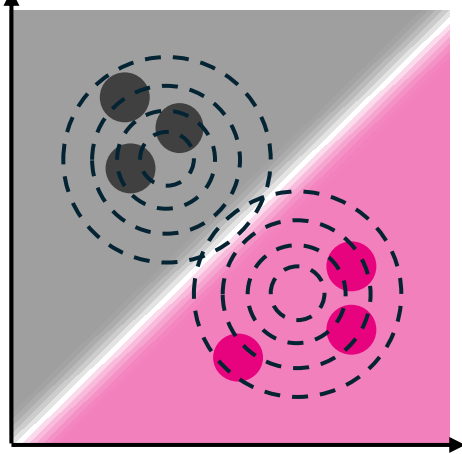
Semantic “shift”



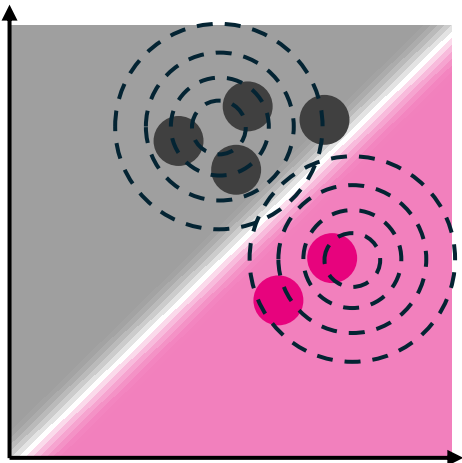
$$p(x)p(y|x) \rightarrow q(x)q(y|x)$$

What do we want to detect?

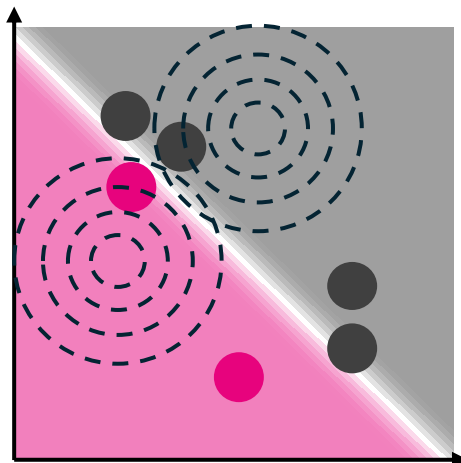
ID



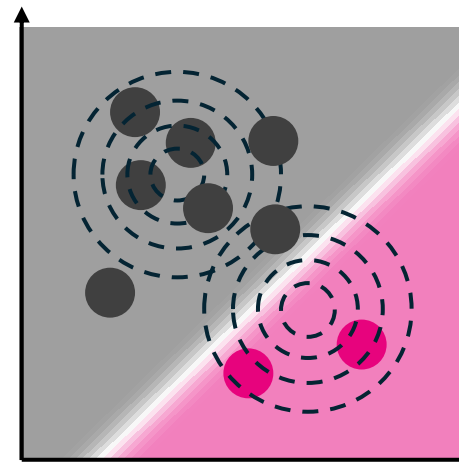
covariate



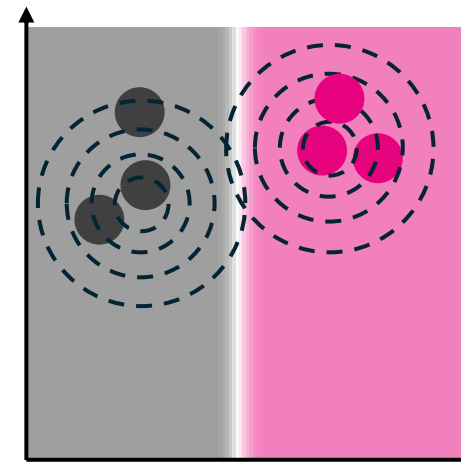
concept



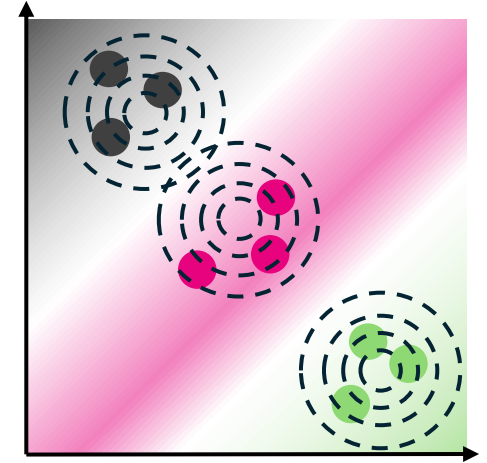
label



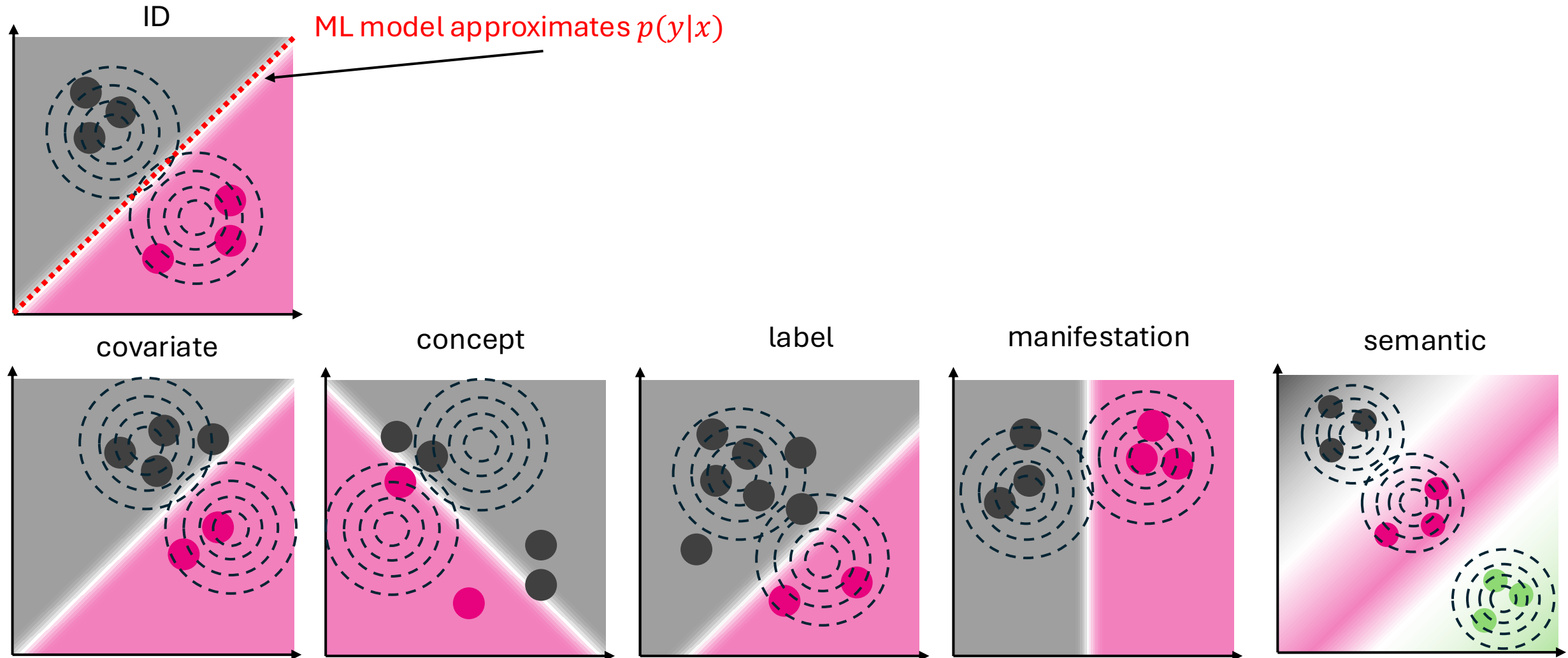
manifestation



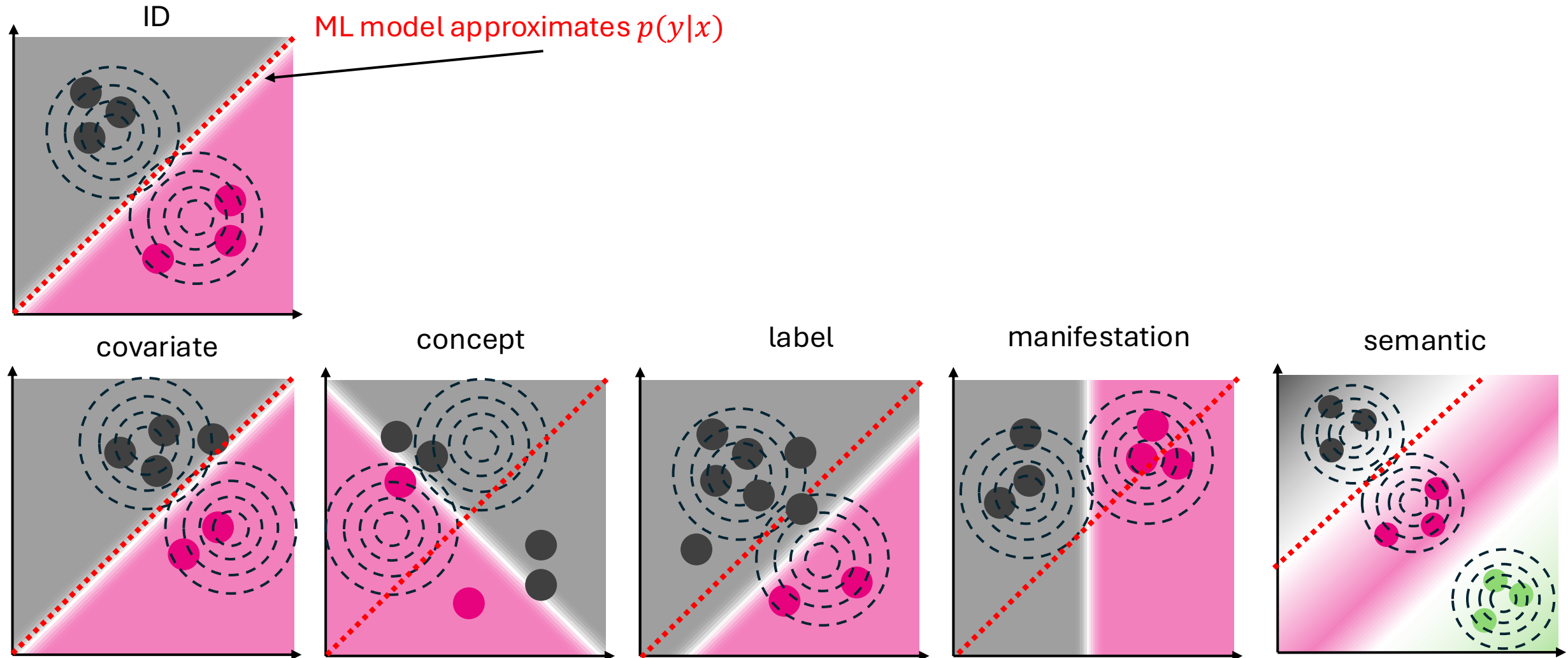
semantic



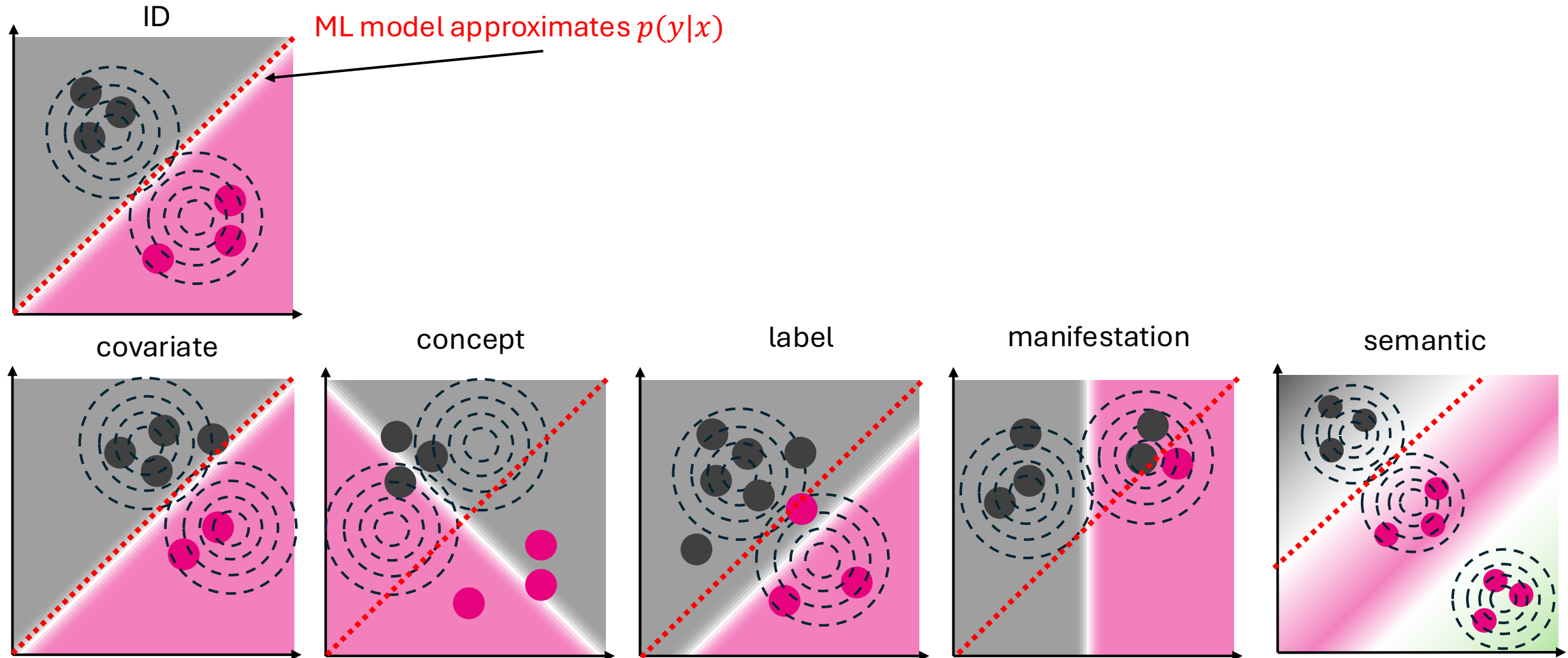
What do we want to detect?



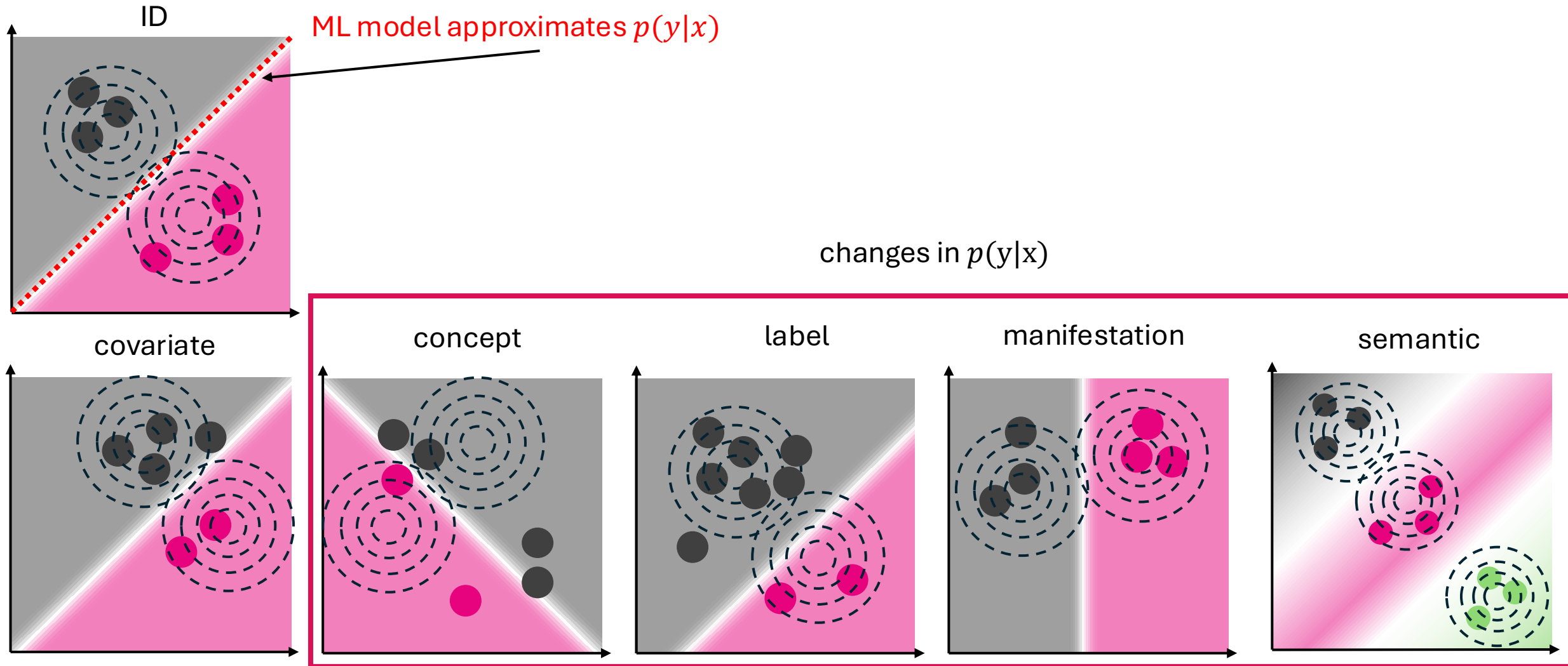
What do we want to detect?



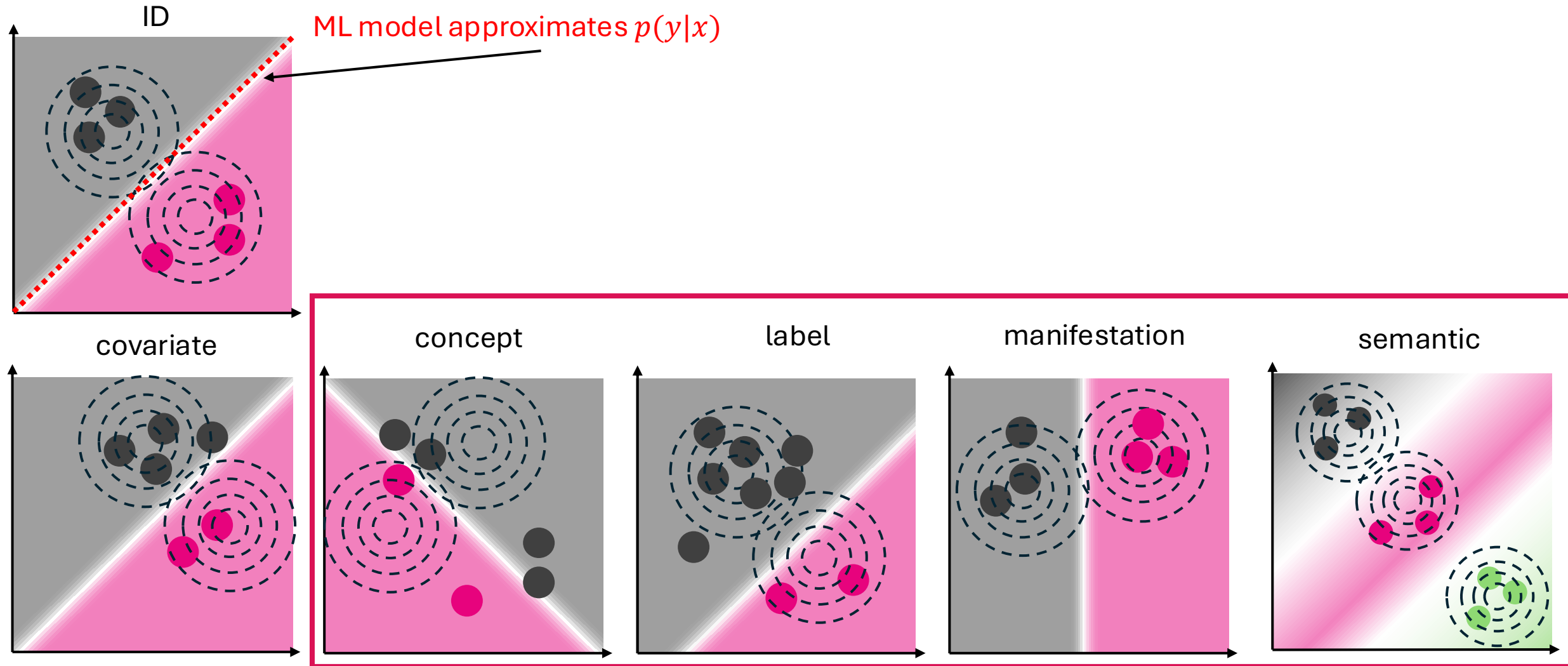
What do we want to detect?



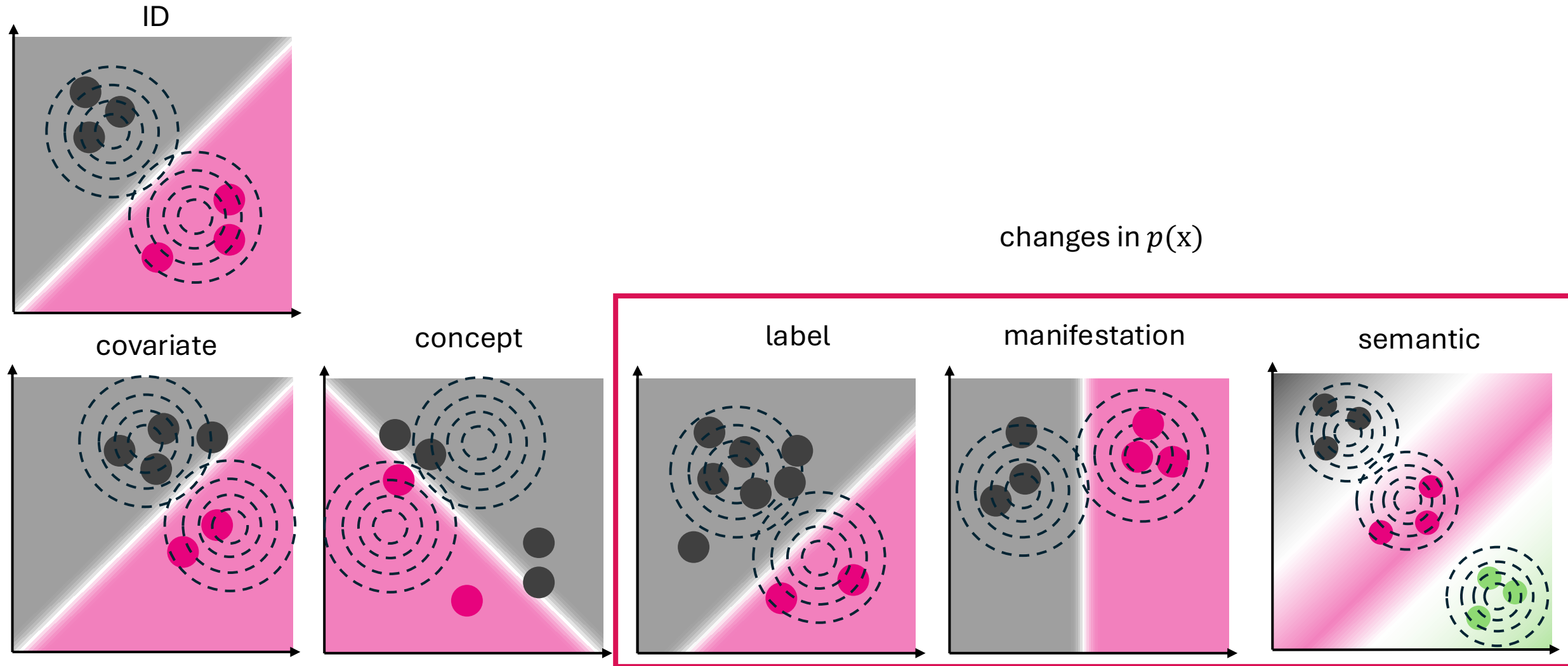
What do we want to detect?



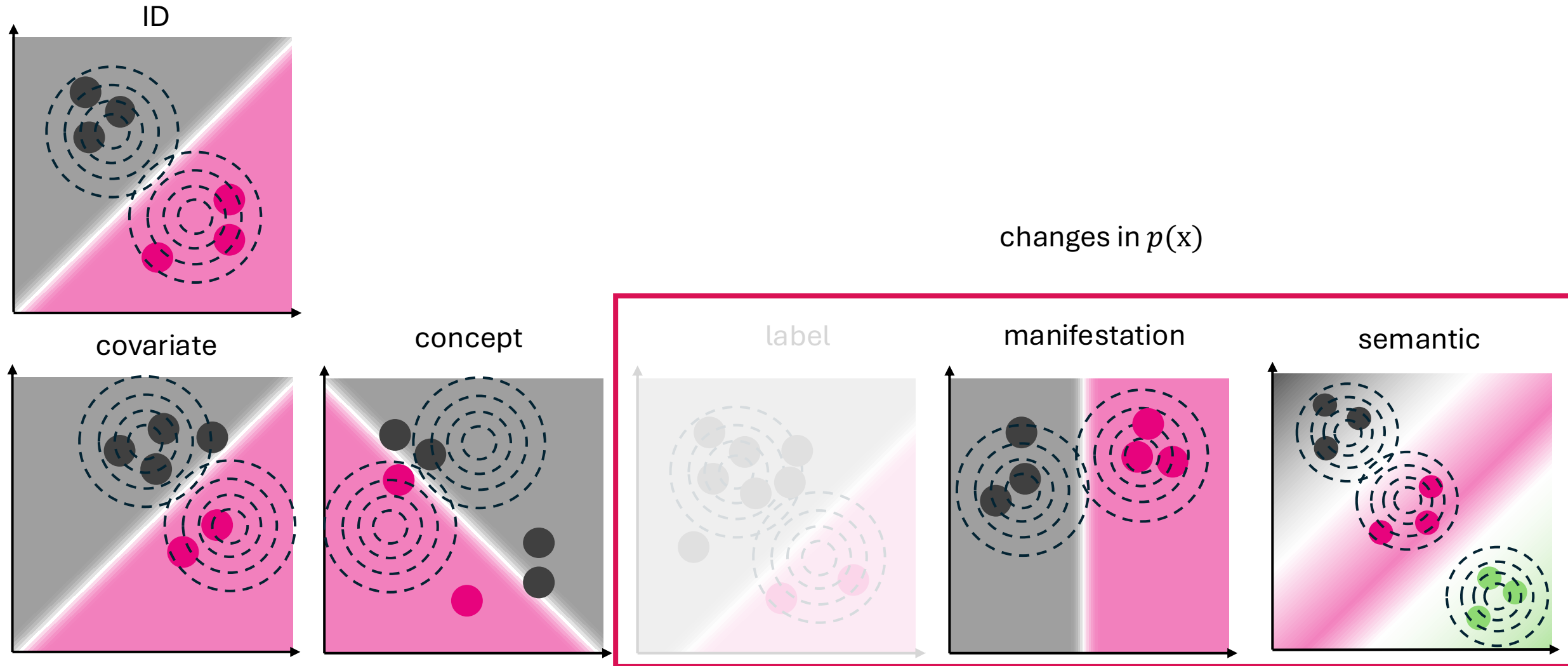
What can we detect?



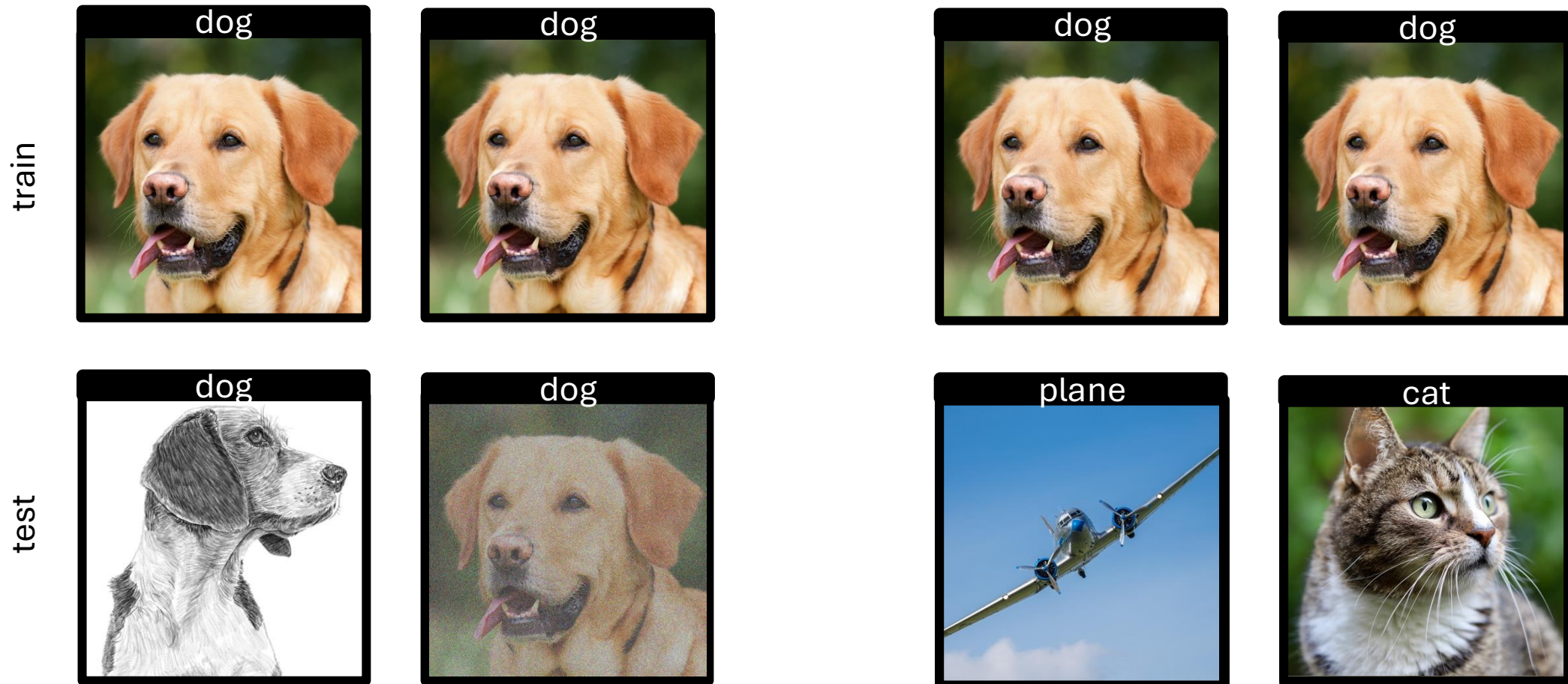
What can we detect?



What can we detect?



From Theory to Reality



From Theory to Reality

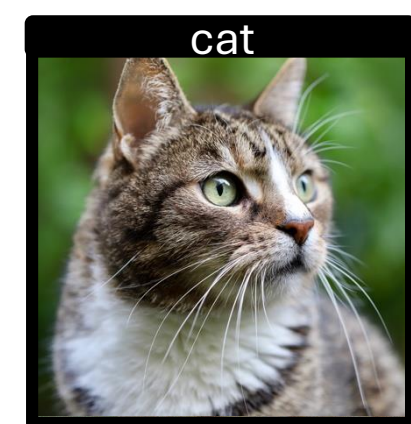
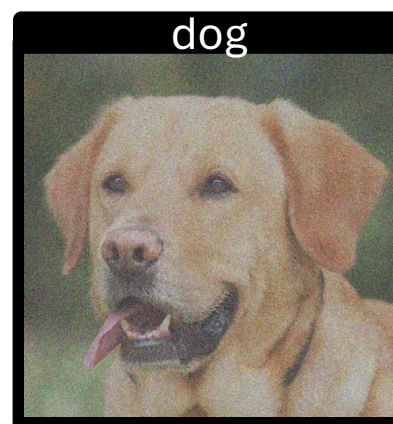
covariate

semantic

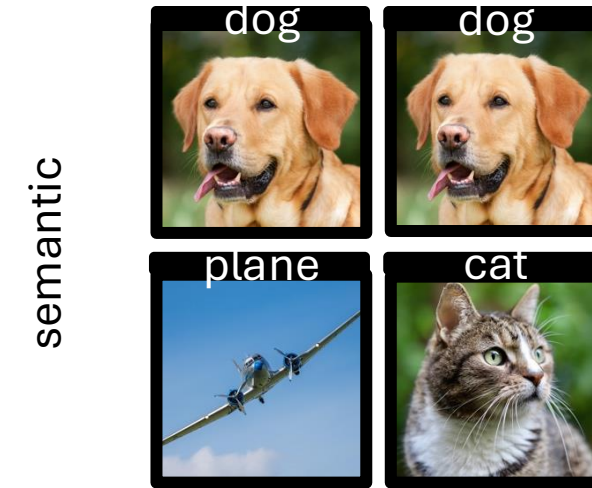
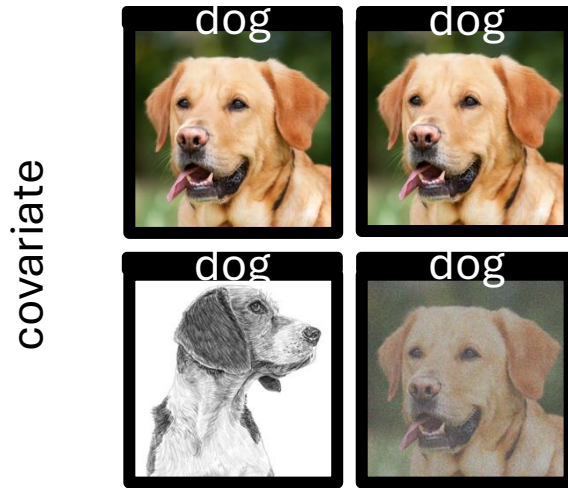
train



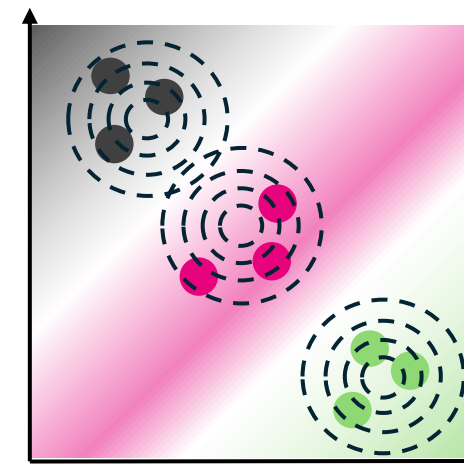
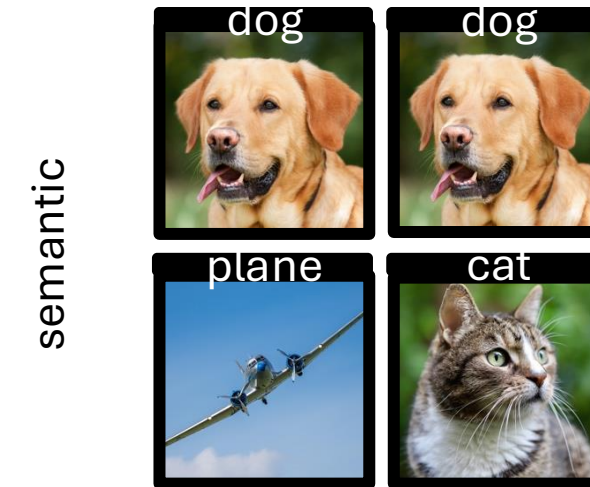
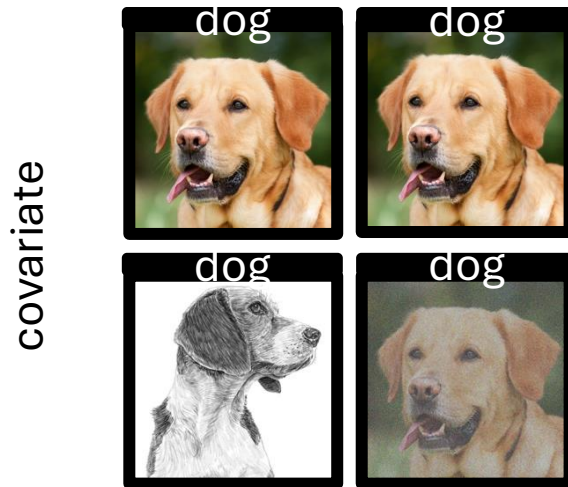
test



From Theory to Reality ... and back



From Theory to Reality ... and back

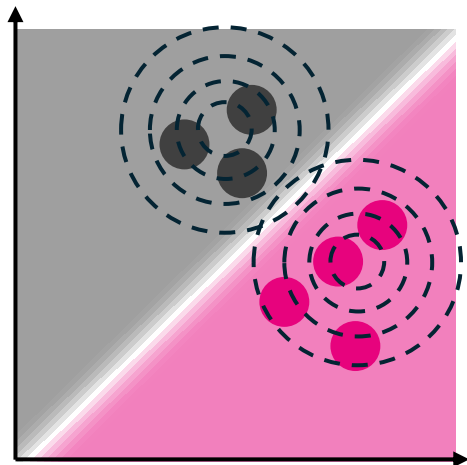
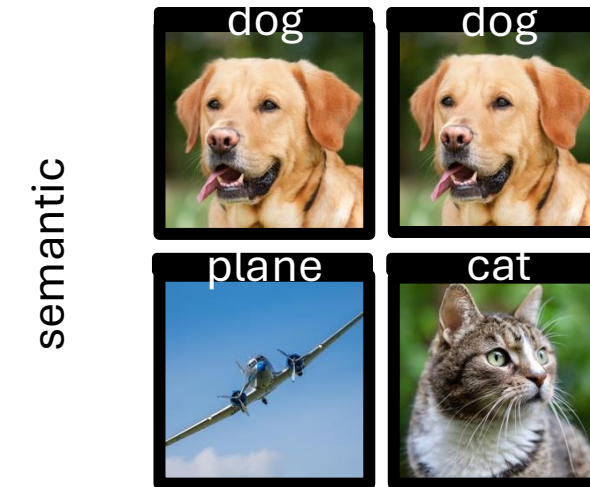
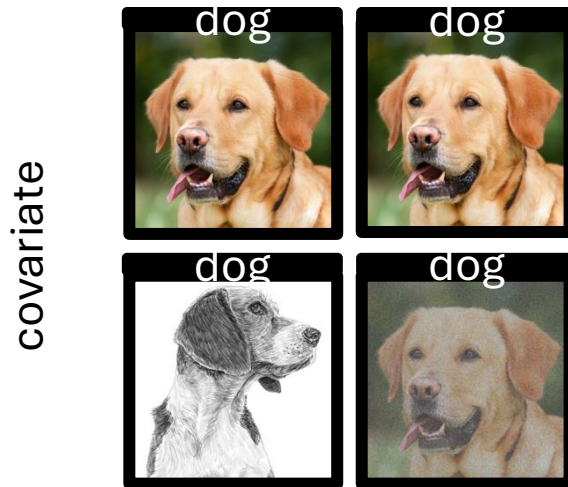


semantic

$$p(x)p(y|x) \rightarrow q(x)q(y|x)$$

Changed x **and** y

From Theory to Reality ... and back

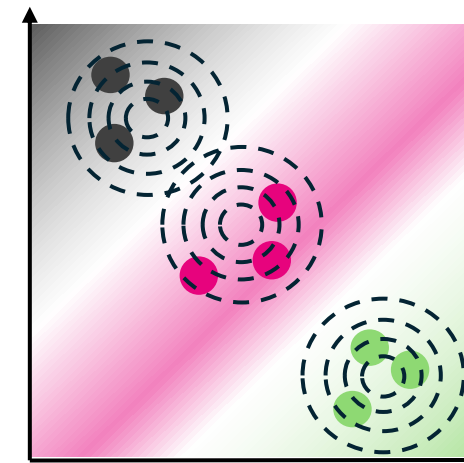


covariate

$$p(x)p(y|x) \rightarrow q(x)p(y|x)$$

$$q(y) = \int p(y|x) \cdot q(x) dx$$

$$q(x|y) = \frac{p(y|x)q(x)}{q(y)}$$

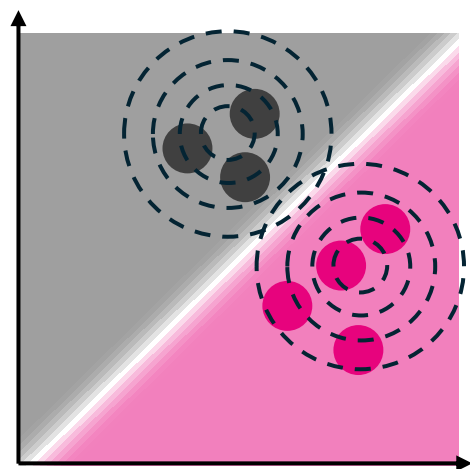
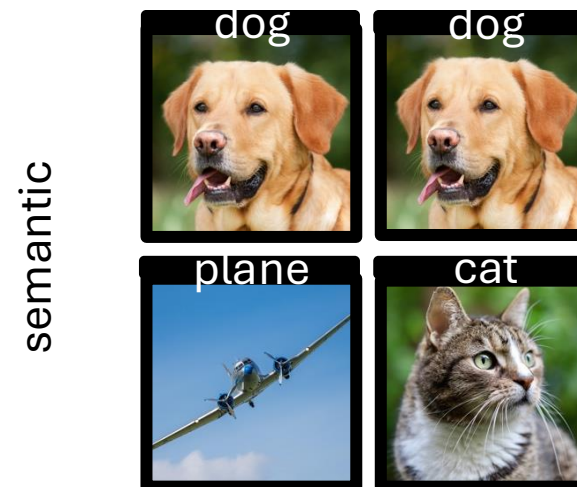
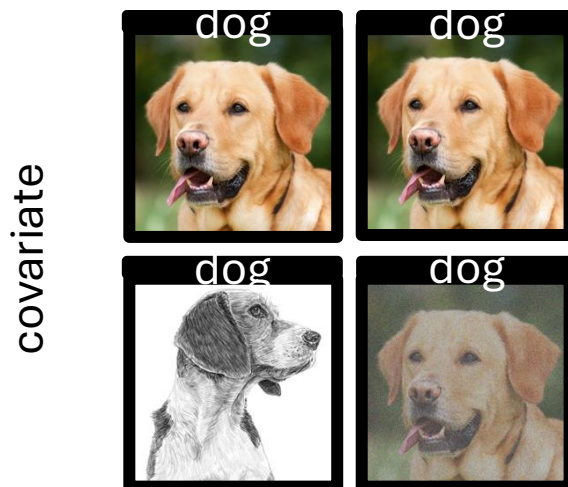


semantic

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Changed x **and** y

From Theory to Reality ... and back

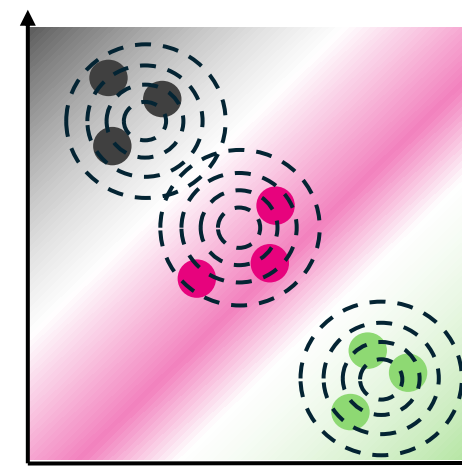


covariate

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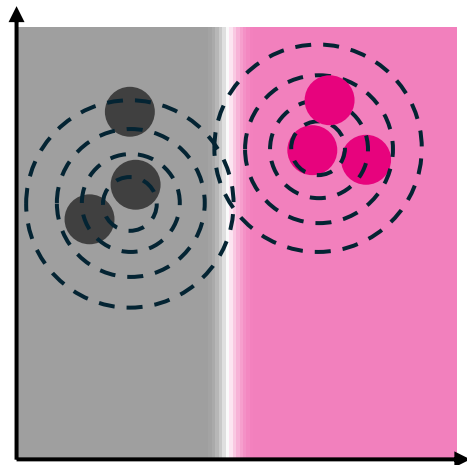
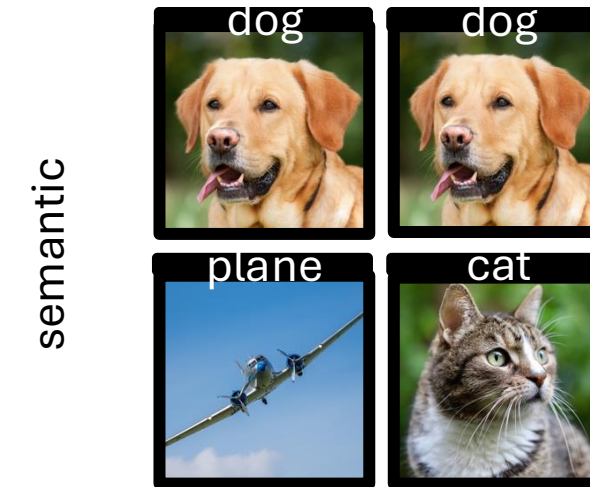
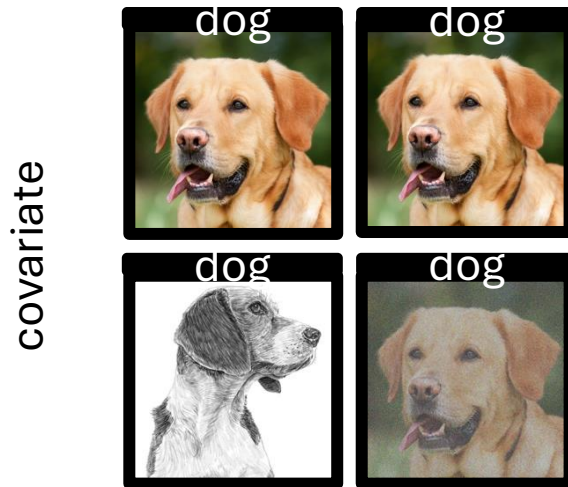


semantic

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Changed x **and** y

From Theory to Reality ... and back



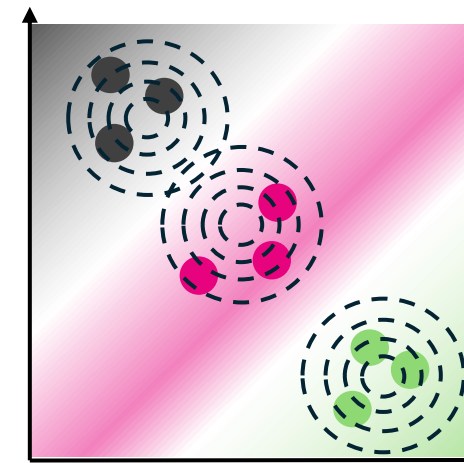
manifestation

$$p(y)\mathbf{p}(x|y) \rightarrow p(y)\mathbf{q}(x|y)$$

$$q(x) = \int q(x|y) \cdot p(y) dy$$

$$q(y|x) = \frac{q(x|y)p(y)}{q(x)}$$

Y presents itself differently in X



semantic

$$p(x)p(y|x) \rightarrow q(x)q(y|x)$$

Changed x **and** y

Fields acknowledging the open world

Fields acknowledging the open world

- OOD detection
- Open set recognition
- Anomaly detection
- Novelty detection
 - one-class / multi-class
- Outlier detection

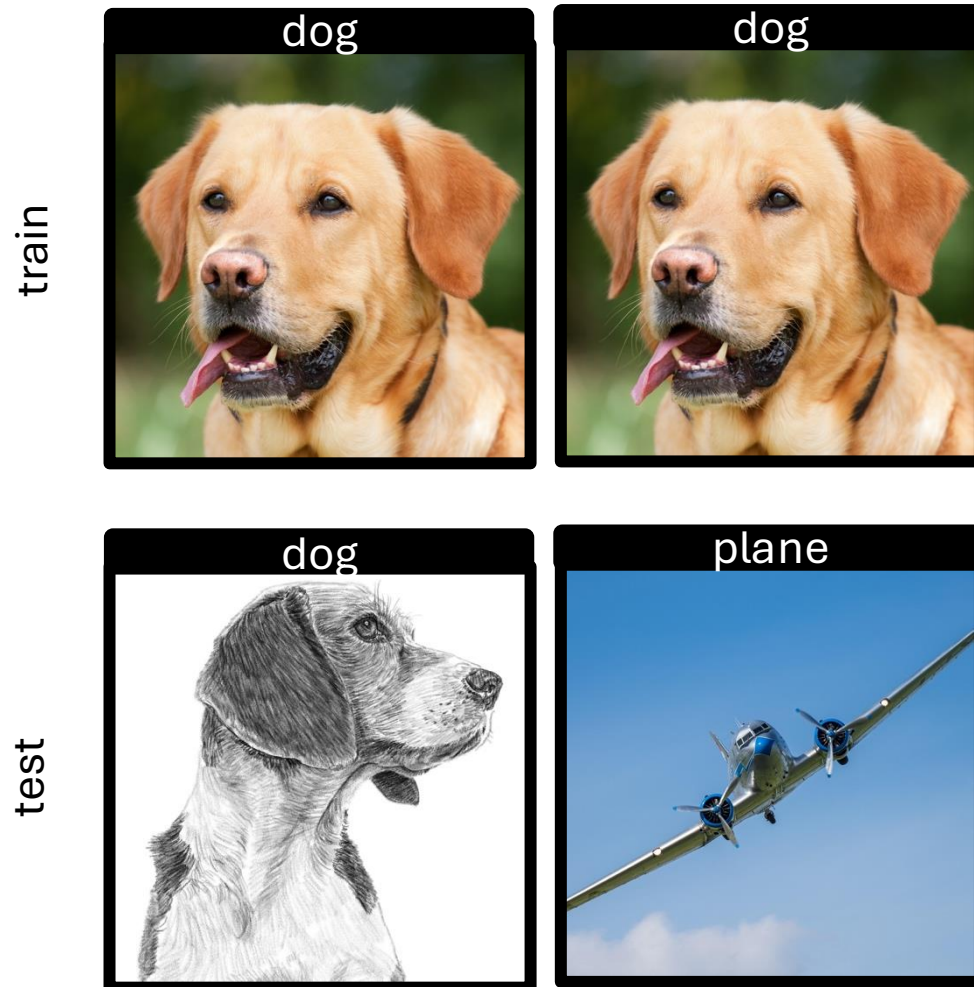
Fields acknowledging the open world

- OOD detection
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- Novelty detection
 - one-class / multi-class
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- Learning with rejection
- Generalisation
- Novelty discovery
- Zero-shot learning
- Open-world recognition

Fields acknowledging the open world

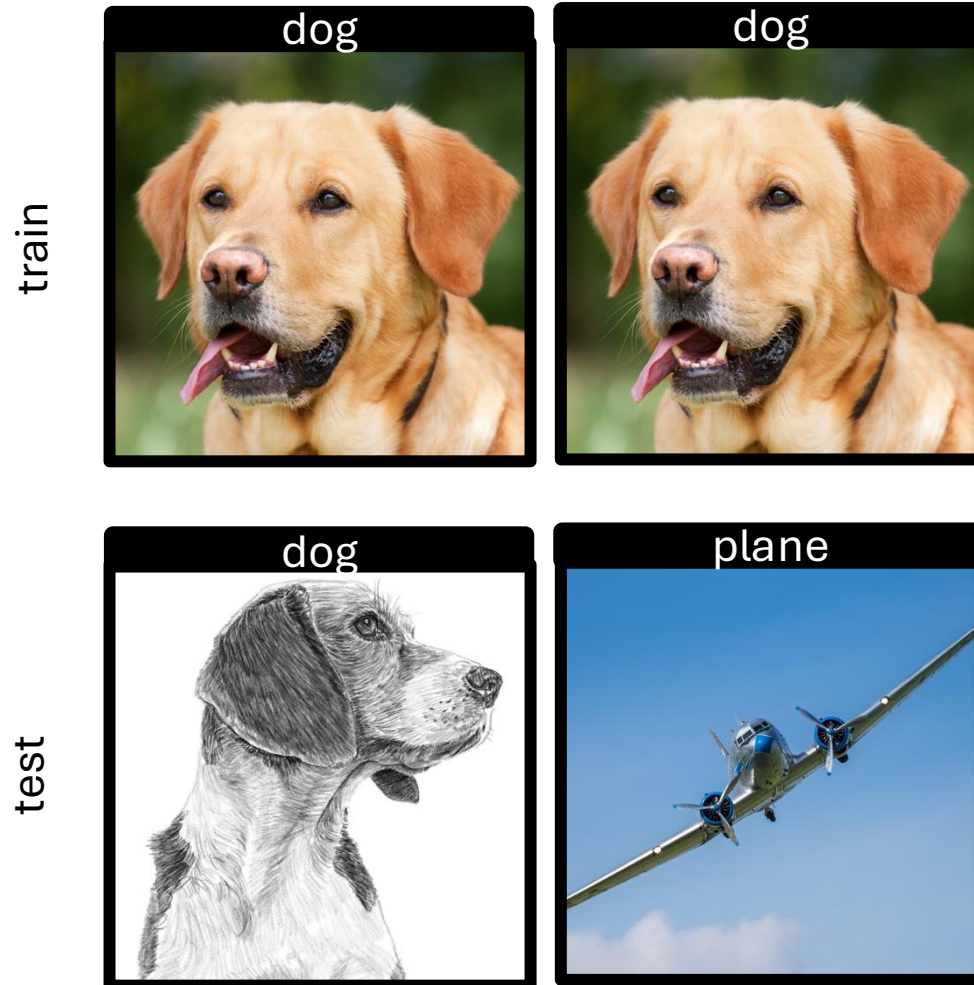
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Detecting OOD: Near and Far



Dong et al. (2022) *CVPR*
Fort et al. (2021) *NeurIPS*
Mukhoti et al. (2023) *ICCVW*
Winkens et al. (2020) *arXiv*
Yang et al. (2022) *NeurIPS*
Yang et al. (2023) *International Journal of Computer Vision*
Yang et al. (2025) *ICLR*

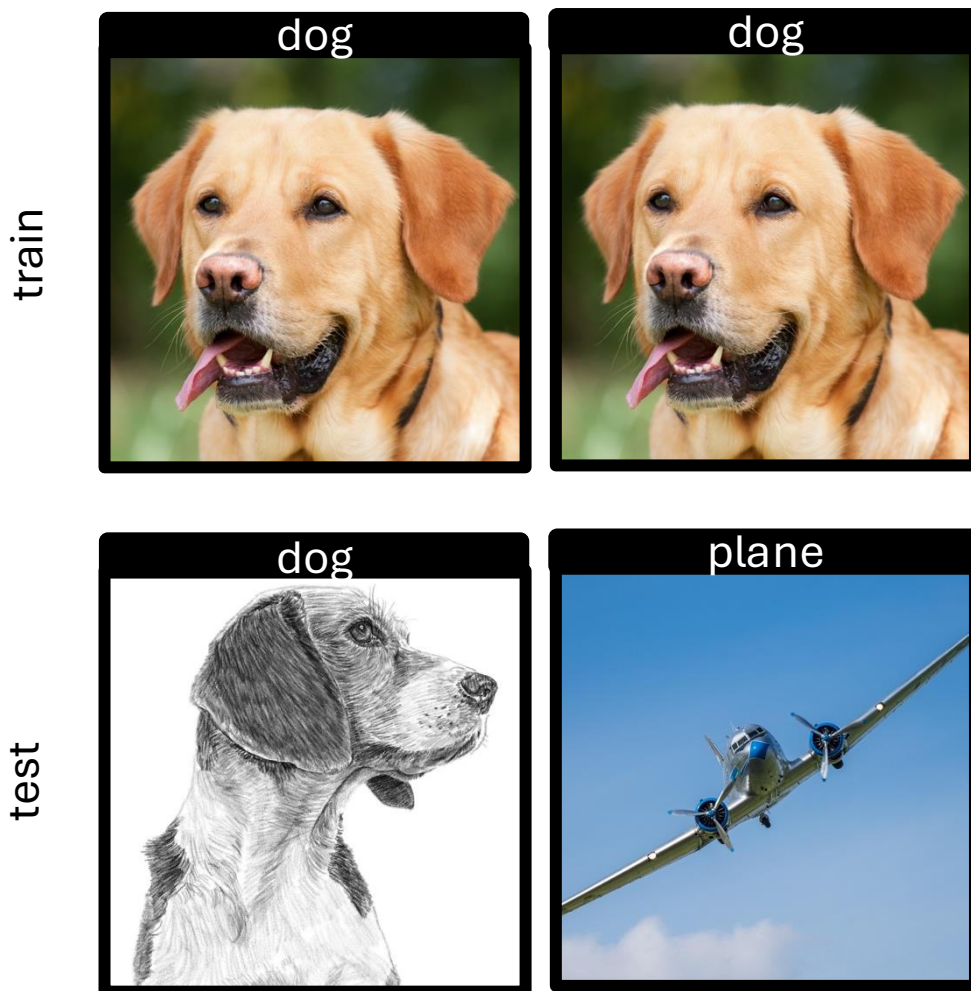
Detecting OOD: Near and Far



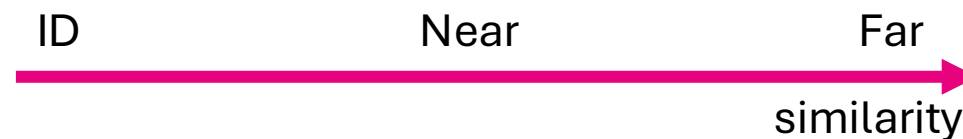
- No consensus definition in literature

Dong et al. (2022) *CVPR*
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Detecting OOD: Near and Far



- No consensus definition in literature



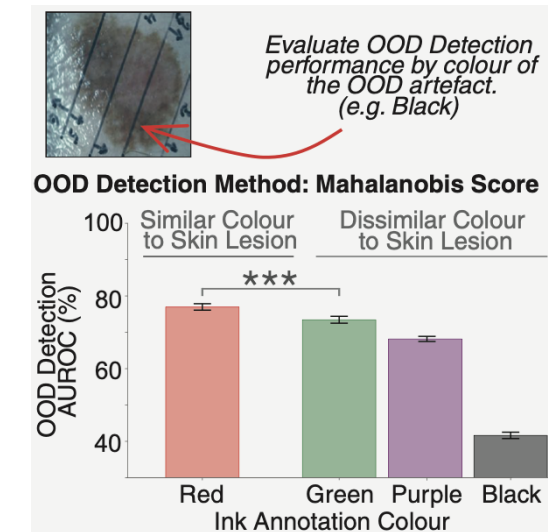
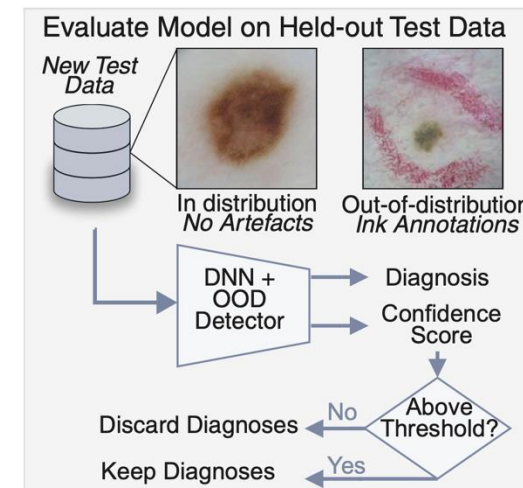
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OOD Detection Methods

Classification	Output-based Methods	Training-free, post-hoc
		Training-based
	Outlier Exposure	Real Outlier
		Data Generation
	Gradient-based methods	
	Bayesian Models	
	OOD for Foundation Models	
Density-based Methods		
Distance-based Methods		
Reconstruction-based Methods		

Methodological limitations

- Most methods are more sensitive to covariate (manifestation) shift than semantic shift
- Benchmarks
 - Unrealistic and/or small (ImageNet, CIFAR)
 - Overlapping classes
- “Invisible gorilla effect”



Anthony et al. (2026) *CVPR*
Humblot-Renaux (2026)
Mukhoti et al. (2023) *ICML*
Yang et al. (2024) *ICLR*

Yang et al. (2024) *International Journal of Computer Vision*

Conceptual limitations

- $p_{train}(x, y)$ (ID) is only vaguely defined
- Tail or OOD?
- Generalisation or OOD?
- Some go as far as to argue using supervised models trained on in-distribution data for semantic shift detection is fundamentally flawed
- ID does not consequently mean correct prediction

Ahmed et al. (2020) AAAI

Guerin et al. (2023) AAAI

Li et al. (2025) ICML

Yang et al. (2023) *International Journal of Computer Vision*

Instead: Out of model scope

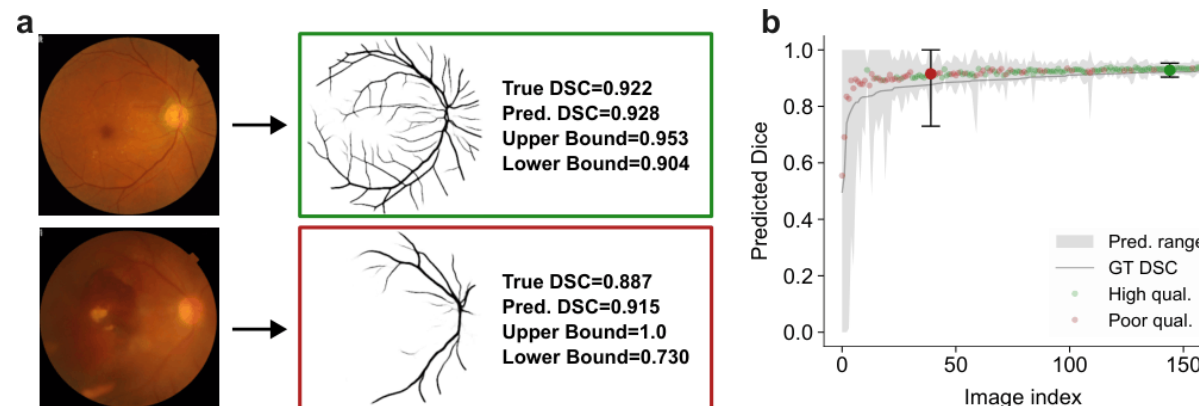
- Discard likely erroneous predictions
- E.g. performance prediction

Instead: Out of model scope

- Discard likely erroneous predictions
- E.g. performance prediction

Conformal Performance Range Prediction for Segmentation Output Quality Control

Anna M. Wundram¹, Paul Fischer^{1,5}, Michael Mühlebach², Lisa M. Koch^{3,4},
and Christian F. Baumgartner^{1,5}



Averly et al.(2023) ICCV
Guérin et al. (2023) AAAI
Wundram et al. (2024) UNSURE @ MICCAI

Take-aways

- OOD implies $p_{train}(x, y) \neq p_{test}(x, y)$
- Reasons for this difference include shifts in the joint distribution
- Multiple fields deal with OOD inputs, one of them OOD detection
- But the joint distribution is only vaguely defined, so how can we detect deviations from it?
- Instead, move to out-of-model-scope detection

Hands-on

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